Triple quantum K-theory of flag varieties

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Abstract

We prove a triple quantum K-Pieri formula expanding the product of a quantum double Grothendieck polynomial and a quantized top block. Its coefficients are indexed by endpoints of admissible chains in the quantum Bruhat graph. A cyclic rotation of values turns these chains into the Pieri chains of Lenart and Sottile in the classical k-Bruhat order; this gives uniqueness of the chain to each endpoint and a closed formula for its quantum degree. We prove the formula for $k = 1$, at $y = 0$ for all k, and on the hyperplane $z = 1/\beta$ for all k, where it is an equivariant quantum K-Chevalley formula for the line bundle $\mathcal{O}(-\varpi_k)$ indexed by chains; the general equivariant case is reduced to a junction identity whose remaining part is stated precisely.

1 Polynomial conventions

Let S_∞ denote the permutations of the positive integers with finite support. We use one-line notation, let $\ell(w)$ be the inversion number of w, and write t_{ab} for the transposition interchanging positions $a < b$. Thus wt_{ab} is obtained from w by interchanging $w(a)$ and $w(b)$.

Let β, z, and the entries of $x = (x_1, x_2, \ldots)$, $y = (y_1, y_2, \ldots)$, and $q = (q_1, q_2, \ldots)$ be independent indeterminates. Put

$$X_i = 1 + \beta x_i, \qquad Q_i = \beta^2 q_i, \qquad \overline{y}_a = -\frac{y_a}{1 + \beta y_a}.$$

Our formal group law is $a \oplus b = a + b + \beta ab$, so that $a \oplus \overline{a} = 0$. In particular, $z + \overline{y}_a$ below denotes an *ordinary* sum, not $z \oplus \overline{y}_a$.

We normalize the double Grothendieck polynomials by

$$\mathfrak{G}_{w_0^{(n)}}(x; y) = \prod_{\substack{i,j \geq 1 \\ i+j \leq n}} (x_i \oplus y_j),$$

and the isobaric divided-difference recursion

$$\mathfrak{G}_{ws_i}(x; y) = \pi_i^\beta \mathfrak{G}_w(x; y) \quad \text{if } \ell(ws_i) < \ell(w), \qquad \pi_i^\beta f = \frac{(1 + \beta x_{i+1}) f - (1 + \beta x_i) s_i f}{x_i - x_{i+1}}.$$

Here s_i interchanges x_i, x_{i+1} and fixes all other variables. In this convention $\mathfrak{G}_{21} = x_1 + y_1 + \beta x_1 y_1$, and $\mathfrak{G}_w|_{\beta=0}$ is the double Schubert polynomial with second alphabet $-y$.

For $0 \leq r \leq j$ define

$$F_r^j(X; q, \beta) = \sum_{\substack{J \subseteq \{1, \ldots, j\} \\ |J| = r}} \left(\prod_{a \in J} X_a \right) \left(\prod_{\substack{a \in J \\ a+1 \notin J}} (1 - \beta^2 q_a) \right). \tag{1}$$

In particular $F_0^j = 1$. The Lenart–Maeno quantization map $\mathcal{Q}$ is linear over the coefficient parameters and is specified on standard elementary monomials by

$$\mathcal{Q}\left(\prod_{j=1}^{N-1} e_{r_j}(X_1,\dots,X_j)\right) = \prod_{j=1}^{N-1} F_{r_j}^{j}(X;q,\beta), \qquad 0 \le r_j \le j. \tag{2}$$

One takes N sufficiently large for the polynomial being quantized. The standard elementary monomials form a basis of the space with $\deg_{X_i} \le N-i$, $1 \le i < N$; this specifies the map on that space. The map is not asserted to be an algebra homomorphism. We set

$$\mathfrak{G}_u^q(x;y) = \mathcal{Q}\big(\mathfrak{G}_u(x;y)\big).$$

At $\beta = -1$ this is the usual quantum double Grothendieck normalization associated with Lenart–Maeno quantization [5, 4].

The top block appearing in our formula is

$$B_k^q(x;z) := \mathcal{Q}\left(\prod_{i=1}^{k}(x_i+z)\right) = \beta^{-k}\sum_{r=0}^{k}\big(-(1-\beta z)\big)^{k-r} F_r^k(X;q,\beta). \tag{3}$$

The apparent negative powers of β in this expression cancel. At $\beta = 0$ the block is understood by polynomial specialization, not by separately substituting into β^{-k}. We set $B_0^q = 1$. Notice that (3) quantizes the ordinary factors $x_i + z$, not the formal-group factors $x_i \oplus z$.

2 Chains and endpoint factors

Fix $u \in S_\infty$ and $k \ge 1$. Choose a finite rank $R > k$ with $u \in S_R$. The directed quantum Bruhat graph on S_R has an edge

$$v \xrightarrow{(a,b)} vt_{ab} \qquad (1 \le a < b \le R)$$

when either

$$\begin{aligned} &\ell(vt_{ab}) = \ell(v) + 1 && \text{(a Bruhat edge)},\\ &\ell(vt_{ab}) = \ell(v) - 2(b-a) + 1 && \text{(a quantum edge)}. \end{aligned}$$

A Bruhat edge has weight 1; a quantum edge labeled (a,b) has weight $q_a q_{a+1}\cdots q_{b-1}$. Order labels by

$$(a,b) \prec (c,d) \quad\Longleftrightarrow\quad b > d \ \text{ or } \ (b = d \text{ and } a < c).$$

Definition 1. *An admissible k-chain from u in S_R is a directed path*

$$\gamma: \quad u = v_0 \xrightarrow{(a_1,b_1)} v_1 \xrightarrow{(a_2,b_2)} \cdots \xrightarrow{(a_s,b_s)} v_s = w$$

in the quantum Bruhat graph satisfying:

1. *$a_i \le k < b_i$ for every i;*
2. *all labels (a_i,b_i) are distinct;*
3. *$b_1 \ge b_2 \ge \cdots \ge b_s$;*

4. if $i < s$ and $a_i = a_j$ for some $j < i$, then $(a_i, b_i) \prec (a_{i+1}, b_{i+1})$.

The empty path, with endpoint u, is included. Let $E_R(u,k)$ be the set of endpoints of these paths. Write $s(\gamma)$ for the number of edges, and define $D(\gamma) = (D_1, \ldots, D_{R-1})$ by

$$D_j(\gamma) = \#\{\, i : \textit{the } i\textit{th edge is quantum and } a_i \le j < b_i \,\}.$$

Thus the weight of γ is $q^{D(\gamma)} = \prod_{j=1}^{R-1} q_j^{D_j(\gamma)}$.

Directly from the edge definitions, every such path satisfies

$$s(\gamma) = \ell(w) - \ell(u) + 2\sum_{j=1}^{R-1} D_j(\gamma). \tag{4}$$

For an endpoint w define three disjoint sets of *values* from the initial window $u(1), \ldots, u(k)$:

$$\begin{aligned}
\mathrm{Fix}(u,w;k) &= \{u(i) : 1 \le i \le k,\ w(i) = u(i)\},\\
\mathrm{Left}(u,w;k) &= \{u(i) : 1 \le i \le k,\ w^{-1}(u(i)) < i\},\\
\mathrm{Out}(u,w;k) &= \{u(i) : 1 \le i \le k\} \setminus \big(\mathrm{Fix}(u,w;k) \cup \mathrm{Left}(u,w;k)\big).
\end{aligned}$$

Set

$$m(u,w;k) = |\,\mathrm{Left}(u,w;k)| + |\,\mathrm{Out}(u,w;k)| = k - |\,\mathrm{Fix}(u,w;k)|.$$

The name Out includes values that move right while remaining in the window, as well as values that leave the window.

3 The triple quantum K-Pieri formula

We state the triple quantum K-Pieri formula, expanding a quantum double Grothendieck polynomial in the y-equivariant basis times a quantized top block carrying a second parameter z.

Theorem 2 (Triple quantum K-Pieri formula)**.** *Let $k \ge 1$ and $u \in S_\infty$. Set*

$$R_0 = \max\{\max(\{2\} \cup \mathrm{supp}(u)),\ k+1\} + k, \qquad E = E_{R_0}(u,k).$$

For each endpoint $w \in E$, all admissible chains from u to w have the same quantum degree $D(u,w;k)$ and length $s(u,w;k)$. Moreover, $s(u,w;k) \ge m(u,w;k)$, and

$$B_k^q(x;z)\,\mathfrak{G}_u^q(x;y) = \sum_{w \in E} C_{u,w}^{(k)}(y,z,q,\beta)\,\mathfrak{G}_w^q(x;y), \tag{5}$$

where

$$\boxed{\begin{aligned} C_{u,w}^{(k)} = q^{D(u,w;k)}\beta^{s(u,w;k)-m(u,w;k)} &\prod_{a \in \mathrm{Fix}(u,w;k)} \left(z - \frac{y_a}{1+\beta y_a}\right)\\ &\times (1-\beta z)^{|\,\mathrm{Left}(u,w;k)|} \prod_{a \in \mathrm{Out}(u,w;k)} (1+\beta y_a)^{-1}. \end{aligned}} \tag{6}$$

Here $B_k^q(x;z) = \mathcal{Q}(\prod_{i=1}^k (x_i + z))$ is defined in (3); D records the quantum edges in a chain; s is its number of edges; and $m = |\,\mathrm{Left}\,| + |\,\mathrm{Out}\,| = k - |\,\mathrm{Fix}\,|$. The sets $\mathrm{Fix}, \mathrm{Left}, \mathrm{Out}$ record whether each value among $u(1), \ldots, u(k)$ stays fixed, moves to an earlier position, or otherwise moves (to a later position or out of the first k positions). The admissible chains and their quantum weights are defined above.

The formula is indexed by endpoints, so each endpoint appears once. The chain-independence of D and s is part of the theorem; it is proved in Corollary 9, which shows that there is exactly one admissible chain to each endpoint and gives D and s in closed form, see (7).

Remark 3 (Rank and polynomial setting)**.** *The rank R_0 just makes the endpoint set finite; it does not impose relations from a finite flag variety. For $k = 0$ the formula reduces to $\mathfrak{G}_u^q = \mathfrak{G}_u^q$.*

Remark 4 (A useful degree check)**.** *Let $|D| = \sum_j D_j$. By (4), the prefactor in (6) can also be written as*

$$q^D \beta^{s-m} = Q^D \beta^{\ell(w)-\ell(u)-m}.$$

The first expression is preferable for specialization at $\beta = 0$, since the inequality $s \geq m$ makes nonnegativity of the power explicit. With $\deg x_i = \deg y_i = \deg z = 1$, $\deg \beta = -1$, and $\deg q_i = 2$, the coefficient has degree $k + \ell(u) - \ell(w)$.

4 Proof calculations

The arguments below establish the rank bound, the uniqueness of the admissible chain to each endpoint together with closed formulas for s and D (by a rotation of values to the classical k-Bruhat order), polynomiality of the top block, the full formula for $k = 1$, the full formula at $y = 0$ for all k (Theorem 18, from the Naito–Sagaki marked-chain formula), and the full equivariant formula at $z = 1/\beta$ for all k (Theorem 22, by the junction identity). The remaining step for the general equivariant formula is specified at the end of the section; that step is not proved here.

4.1 Finiteness and the power of β

Proposition 5. *Put $n(u) = \max(\{1\} \cup \operatorname{supp}(u))$ and $M = \max\{n(u), k\} + 1$. Every label in an admissible k-chain from u has second entry at most M. Consequently its endpoint set stabilizes in every rank containing u and $\{1, \ldots, M\}$. For every such chain γ with endpoint w,*

$$s(\gamma) \geq m(u, w; k).$$

Proof. The first label has maximal second entry, by the ordering condition. Suppose it were (a, b) with $b > M$. Then $a \leq k < b - 1$, $u(b) = b$, $u(b-1) = b-1$, and $u(a) < b - 1$. The intermediate value $u(b-1)$ prevents (a, b) from giving a Bruhat cover. Since $u(a) < u(b)$, the edge cannot be quantum either. This contradiction proves the rank bound. Adding fixed positions after M changes neither the relevant inversion differences nor the permitted paths, proving stabilization.

Every position $i \leq k$ with $w(i) \neq u(i)$ must occur in at least one transposition of the path. Each transposition involves exactly one position in $\{1, \ldots, k\}$. There are $k - |\operatorname{Fix}(u, w; k)| = m(u, w; k)$ changed positions, so at least this many edges are required. □

This proves that the displayed power β^{s-m} is nonnegative for each path, without presupposing independence of the path. Independence of s and D for a fixed endpoint, and in fact uniqueness of the path, is proved next.

4.2 Rotation to the classical k-Bruhat order

The quantum Bruhat graph has a cyclic symmetry, which we derive from the following reformulation of the edge condition. For $v \in S_R$ and $1 \leq a < b \leq R$ put

$$A(v; a, b) = \{v(r) : a < r < b\},$$

the set of values at the positions strictly between a and b. For distinct $x, y \in \{1, \dots, R\}$ let $\operatorname{arc}(x, y)$ be the set of values met strictly between x and y when moving upward cyclically from x to y:

$$\operatorname{arc}(x,y) = \begin{cases} \{x+1, \dots, y-1\} & x < y, \\ \{x+1, \dots, R\} \cup \{1, \dots, y-1\} & x > y. \end{cases}$$

Lemma 6 (Cyclic form of the edge condition). *Let $v \in S_R$ and $a < b$. Then $v \xrightarrow{(a,b)} vt_{ab}$ is an edge of the quantum Bruhat graph if and only if*

$$A(v; a, b) \cap \operatorname{arc}(v(a), v(b)) = \varnothing.$$

The edge is a Bruhat edge if $v(a) < v(b)$ and a quantum edge if $v(a) > v(b)$.

Proof. Passing from v to vt_{ab} changes the inversion status of the pair of positions (a, b) and of the pairs (a, r), (r, b) with $a < r < b$ exactly when $v(r)$ lies strictly between $v(a)$ and $v(b)$; no other pair is affected. Hence, if $v(a) < v(b)$,

$$\ell(vt_{ab}) = \ell(v) + 1 + 2\,\#\{r : a < r < b,\ v(a) < v(r) < v(b)\},$$

and if $v(a) > v(b)$,

$$\ell(vt_{ab}) = \ell(v) - 1 - 2\,\#\{r : a < r < b,\ v(b) < v(r) < v(a)\}.$$

A Bruhat edge therefore means $v(a) < v(b)$ and $A(v; a, b) \cap (v(a), v(b)) = \varnothing$. A quantum edge means $v(a) > v(b)$ and that all $b - a - 1$ intermediate values lie in $(v(b), v(a))$; since $A(v; a, b)$ does not contain $v(a)$ or $v(b)$, this says that $A(v; a, b)$ avoids the complement of $[v(b), v(a)]$, which is $\operatorname{arc}(v(a), v(b))$. □

For $0 \le \vartheta \le R$ let $\rho = \rho_\vartheta$ be the cyclic shift of $\{1, \dots, R\}$ given by

$$\rho(x) = x - \vartheta \quad (x > \vartheta), \qquad \rho(x) = x - \vartheta + R \quad (x \le \vartheta),$$

so that $\rho(\vartheta) = R$ is the largest value when $\vartheta \ge 1$, and the relabelled order of values is $\vartheta+1 < \vartheta+2 < \cdots < R < 1 < \cdots < \vartheta$. For $v \in S_R$ write $\rho v = \rho \circ v$ for the permutation obtained by relabelling the values of v. Since ρ is a cyclic shift, $\rho(\operatorname{arc}(x, y)) = \operatorname{arc}(\rho x, \rho y)$, and clearly $A(\rho v; a, b) = \rho(A(v; a, b))$. Lemma 6 gives at once:

Corollary 7 (Cyclic symmetry). *(a, b) is an edge from v if and only if it is an edge from ρv. The two edges have the same type unless exactly one of $v(a), v(b)$ is at most ϑ: the edge from v is quantum and the edge from ρv is Bruhat exactly when $v(b) \le \vartheta < v(a)$, and the edge from v is Bruhat and the edge from ρv is quantum exactly when $v(a) \le \vartheta < v(b)$.*

The point of the following lemma is that, for the paths we consider, a single rotation removes all quantum edges at once. It uses only the distinctness of the labels and the weak decrease of the upper indices, not condition 4 of Definition 1.

Lemma 8 (Rotation lemma). *Let $u \in S_R$, $1 \le k < R$, $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$. Let*

$$u = v_0 \xrightarrow{(a_1,b_1)} v_1 \xrightarrow{(a_2,b_2)} \cdots \xrightarrow{(a_s,b_s)} v_s$$

be a directed path in the quantum Bruhat graph whose labels satisfy $a_i \le k < b_i$, are pairwise distinct, and have $b_1 \ge b_2 \ge \cdots \ge b_s$. Then every edge of the rotated path $\rho v_0 \to \rho v_1 \to \cdots \to \rho v_s$ is a Bruhat edge. Consequently the ith edge of the original path is quantum if and only if $v_{i-1}(b_i) \le \vartheta < v_{i-1}(a_i)$.

Proof. Put $v'_i = \rho v_i$ and $N = R = \rho(\vartheta)$, the largest value; N sits at position $k+1$ in v'_0. By Corollary 7 the rotated path is a directed path with the same labels, and the last assertion follows from the first by the same corollary. We show that no rotated edge is quantum.

Position $k+1$ is involved in a label (a,b) only when $b = k+1$. Let $(c_1, k+1), \dots, (c_m, k+1)$ be the labels with upper index $k+1$, in order; by the weak decrease of upper indices they are the last m labels of the path, and by distinctness the c_l are pairwise distinct. Thus N occupies position $k+1$ until the label $(c_1, k+1)$ is applied, and position c_1 from then on.

Consider first an edge with label (a,b), $b > k+1$. Then $a < k+1 < b$, so $N \in A(v'; a, b)$ for the permutation v' at which it is applied. A quantum edge from v' would have $A(v'; a, b) \subseteq (v'(b), v'(a))$, which cannot contain the largest value. Hence the edge is Bruhat. The edge $(c_1, k+1)$ is Bruhat because the value N at position $k+1$ exceeds the value at c_1.

It remains to treat $(c_l, k+1)$ for $l \geq 2$. We argue by induction on l, assuming that $(c_1, k+1), \dots, (c_{l-1}, k+1)$ are Bruhat edges of the rotated path. Put $e_0 = N$ and let e_h be the value at position c_h immediately before the edge $(c_h, k+1)$, for $1 \leq h \leq l$. Immediately before $(c_h, k+1)$ the value at position $k+1$ is e_{h-1}, so the Bruhat property of these edges gives

$$N = e_0 > e_1 > \cdots > e_{l-1},$$

and after $(c_h, k+1)$ the value e_{h-1} sits at position c_h, where it stays, since c_h is not touched again. Position c_l is touched by none of the labels $(c_h, k+1)$, $h < l$, so $x := e_l$ has been the value at position c_l throughout. For $h < l$ with $c_h < c_l$ the position c_l lies strictly between c_h and $k+1$, and the Bruhat condition for the edge $(c_h, k+1)$ gives $x \notin (e_h, e_{h-1})$.

Suppose that the edge $(c_l, k+1)$ were quantum. Then $x > e_{l-1}$ and every value at a position strictly between c_l and $k+1$ lies in (e_{l-1}, x); in particular $e_{h-1} < x$ whenever $h < l$ and $c_h > c_l$. Since $e_{l-1} < x < e_0$ and x differs from every e_h, there is a unique $h \in \{1, \dots, l-1\}$ with $e_h < x < e_{h-1}$. If $c_h < c_l$ this contradicts $x \notin (e_h, e_{h-1})$; if $c_h > c_l$ it contradicts $e_{h-1} < x$. Hence the edge is Bruhat. □

A saturated chain in the k-Bruhat order is a path of Bruhat edges with labels (a,b), $a \leq k < b$. Lenart and Sottile [2] consider such chains subject to two conditions, (P1): $b_1 \geq \cdots \geq b_s$, and (P0): if $i < s$ and $a_i = a_j$ for some $j < i$, then $(a_i, b_i) \prec (a_{i+1}, b_{i+1})$, with the same order $\prec$ on labels as above; these are conditions 3 and 4 of Definition 1. Distinctness of labels is automatic for Bruhat chains satisfying (P1), since a repeated label would force equal upper indices in between and a return to a smaller permutation. Their uniqueness theorem [2, Theorem 2.2] states that for given v, w there is at most one saturated chain in the k-Bruhat order from v to w satisfying (P0) and (P1).

Corollary 9 (Classical reduction of the chains)**.** *Let $u \in S_R$, $1 \leq k < R$ with $R \geq \max\{n(u), k\}+1$, $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$. Applying ρ to all vertices is a bijection from the admissible k-chains from u onto the saturated chains in the k-Bruhat order starting at ρu that satisfy (P0) and (P1); labels are preserved. Consequently:*

1. *For every $w \in E(u,k)$ there is exactly one admissible k-chain from u to w. In particular $s(u,w;k)$ and $D(u,w;k)$ are well defined, and the chain is a shortest directed path from u to w in the quantum Bruhat graph.*

2. *Along an admissible chain the value at a position $i \leq k$ changes exactly at the labels with lower index i, and in the rotated order it strictly increases at each of them; the value at a position $b > k$ changes exactly at the labels with upper index b and strictly decreases in the rotated order. In particular $w(i) \neq u(i)$ as soon as i is the lower index of some label, so $m(u,w;k)$ equals the number of distinct lower indices of the chain.*

3. Each position is involved in at most one quantum edge, and

$$D_j(u,w;k)=\begin{cases}\#\{\,i\le j:\ u(i)>u(k+1)\ge w(i)\,\} & j<k,\\ \#\{\,b>j:\ u(b)\le u(k+1)<w(b)\,\} & j\ge k,\end{cases}\qquad s(u,w;k)=\ell(w)-\ell(u)+2\sum_j D_j(u,w;k). \tag{7}$$

Proof. By Lemma 8 the rotation of an admissible k-chain is a path of Bruhat edges with the same labels, hence a saturated chain in the k-Bruhat order, and conditions 3 and 4 are (P1) and (P0). Conversely, by Corollary 7 a chain in the k-Bruhat order from ρu with (P0) and (P1) rotates back to a directed path from u with the same labels, which are distinct by the remark above; so it is an admissible k-chain. The inverse rotation is ρ^{-1}, so this is a bijection.

(1) Two admissible chains from u to w rotate to two chains from ρu to ρw satisfying (P0) and (P1), which coincide by the uniqueness theorem of Lenart and Sottile. Each edge of the quantum Bruhat graph raises the length by at most one, so a directed path from u to w has at least $\ell(\rho w)-\ell(\rho u)$ edges; the rotated chain has exactly this many.

(2) A Bruhat edge (a,b) replaces the value at a by the larger value at b and the value at b by the smaller value at a; no other position changes.

(3) By Lemma 8 the ith edge is quantum exactly when $v_{i-1}(b_i)\le\vartheta<v_{i-1}(a_i)$, that is, when the value entering position a_i is at most ϑ while the value leaving it exceeds ϑ, and the value entering position b_i exceeds ϑ while the value leaving it is at most ϑ. In the rotated order the values $>\vartheta$ precede the values $\le\vartheta$, so by (2) this transition happens at most once at each position, namely when the initial value at a_i exceeds ϑ and the final one does not, and when the initial value at b_i is at most ϑ and the final one exceeds it. A quantum edge (a,b) contributes 1 to D_j exactly when $a\le j<b$. For $j<k$ every quantum edge has $b>k>j$, so D_j counts the quantum edges with $a\le j$, which by the above is the first formula. For $j\ge k$ every quantum edge has $a\le k\le j$, so D_j counts the quantum edges with $b>j$, giving the second formula. The formula for s is (4). □

Remark 10. *The corollary identifies the indexing set of Theorem 2: $w\in E(u,k)$ if and only if ρw is reached from ρu by a Lenart–Sottile chain, that is, $\rho u\xrightarrow{c(k)}\rho w$ in the notation of [2], where ρ rotates the values so that $u(k+1)$ becomes the largest. Thus the quantum endpoint set for u is the classical endpoint set for ρu, and the quantum degree records which window values drop from above $u(k+1)$ to at most $u(k+1)$, and which outside values rise across it. We have checked (7) against the direct enumeration of chains for all $u\in S_n$, $n\le 7$, and all k. Lemma 8 also shows that the chains, but not the line paths of Section 4.5 below, can be analysed in the classical k-Bruhat order; backtracks such as $(k-1,k),(k-1,k)$ are inherently non-classical.*

4.3 An explicit polynomial top block

Write $a=1-\beta z$. For $S\subseteq\{1,\dots,k\}$ with no two consecutive elements, put

$$\epsilon(S)=\begin{cases}1 & k\in S,\\ 0 & k\notin S,\end{cases}\qquad U(S)=\{1,\dots,k\}\setminus\big(S\cup(S+1)\big),\quad S+1=\{j+1:j\in S\}.$$

Proposition 11. *The top block has the polynomial expression*

$$B_k^q(x;z)=\sum_{\substack{S\subseteq\{1,\dots,k\}\\ S\text{ has no consecutive elements}}}(-\beta)^{\epsilon(S)}a^{|S|-\epsilon(S)}\left(\prod_{j\in S}q_jX_j\right)\left(\prod_{i\in U(S)}(x_i+z)\right). \tag{8}$$

In particular, no inverse power of β is needed. For $k \geq 2$ it satisfies

$$\begin{aligned}(1-\beta^2 q_{k-1})B_k^q = \big(x_k + z + \beta(q_{k-1}-q_k) - \beta^2(q_k x_k + z q_{k-1})\big)B_{k-1}^q \\ + (1-\beta z)q_{k-1}(1-\beta^2 q_k)X_k B_{k-2}^q,\end{aligned} \tag{9}$$

with $B_0^q = 1$ and $B_1^q = x_1 + z - \beta q_1 X_1$.

Proof. Multiply (3) by β^k and substitute (1). In each product of factors $1 - Q_j$, choose the term $-Q_j$ for a set S of indices. Such a choice forces $j \in J$ and $j+1 \notin J$ for each $j \in S$, so two elements of S cannot be consecutive. Conversely these are the only restrictions on the remaining choices of J.

Each forced selected position contributes X_j. Each forced unselected position $j+1 \leq k$ contributes $-a$. At every other position, summing the selected and unselected possibilities gives $X_i - a$. Thus the contribution for S to $\beta^k B_k^q$ is

$$(-1)^{|S|} Q^S \left(\prod_{j\in S} X_j\right)(-a)^{|S|-\epsilon(S)} \prod_{i\in U(S)} (X_i - a), \qquad Q^S = \prod_{j\in S} Q_j.$$

Since $|U(S)| = k - 2|S| + \epsilon(S)$, $Q_j = \beta^2 q_j$, and $X_i - a = \beta(x_i + z)$, this expression is β^k times the summand in (8). This proves both that formula and polynomiality.

For the recurrence, introduce the auxiliary polynomials

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z)H_{j-1} + a q_{j-1} X_{j-1} H_{j-2} \quad (j \geq 2).$$

They enumerate tilings of $\{1, \ldots, j\}$ by single positions of weight $x_i + z$ and adjacent pairs $\{i, i+1\}$ of weight $a q_i X_i$. Separating the terms of (8) according to whether $k \in S$ gives

$$B_j^q = H_j - \beta q_j X_j H_{j-1} \qquad (j \geq 1). \tag{10}$$

Set $T_j = (1 - \beta^2 q_j)X_j$. The recurrence for H_j, together with $\beta(x_j + z - \beta q_j X_j) + a = T_j$, gives

$$B_k^q = (x_k + z - \beta q_k X_k)B_{k-1}^q + q_{k-1} X_{k-1} T_k H_{k-2}.$$

For $k \geq 3$, the same calculation at $k-1$ gives

$$\beta B_{k-1}^q = T_{k-1} H_{k-2} - a B_{k-2}^q.$$

For $k = 2$ this last identity follows directly from B_1^q. Multiplying the preceding expression for B_k^q by $1 - \beta^2 q_{k-1}$ and using this identity yields

$$(1-\beta^2 q_{k-1})B_k^q = \frac{T_k - a(1-\beta^2 q_{k-1})}{\beta} B_{k-1}^q + a q_{k-1} T_k B_{k-2}^q.$$

Expanding the numerator gives exactly (9); it is divisible by β. □

At $\beta = 0$, this becomes

$$B_k^q = (x_k + z)B_{k-1}^q + q_{k-1} B_{k-2}^q,$$

the elementary quantum cohomological recurrence. The additional terms in (9) are therefore essential K-theoretic corrections, not changes in notation.

4.4 Passing from finite flags to polynomial identities

We will use the equivariant quantum Chevalley formula, which is initially an identity of classes on a finite flag variety. The following observation explains the passage back to polynomials.

Proposition 12 (Stable detection). *At $\beta = -1$, let P be a polynomial in finitely many x_i and q_i, with coefficients rational in finitely many y_a. If its image in the localized equivariant quantum K-theory presentation is zero in every sufficiently large rank, then $P = 0$ as a polynomial.*

Proof. Suppose $P \neq 0$, and choose its least nonzero total quantum degree. In the quantum-adic expansion of its image, the coefficient of each monomial of that degree is the ordinary equivariant K-theory class of the corresponding coefficient polynomial $f(x; y)$. Indeed, all smaller quantum-degree coefficient polynomials of P are zero. The quantum K-theory module is free on the Schubert classes, so each such ordinary class must vanish.

Let $x_1, \ldots, x_r$ include the variables occurring in f, and let d bound its degree in each of these variables. Choose a rank $N > d + r$ containing all parameters in question. Restricting a vanishing ordinary class to every torus fixed point gives

$$f(\overline{y}_{\sigma(1)}, \ldots, \overline{y}_{\sigma(r)}; y) = 0 \qquad (\sigma \in S_N).$$

Work over the rational function field in the equivariant parameters. The N numbers $\overline{y}_a$ are distinct. After fixing distinct values for $r - 1$ variables, the remaining variable has $N - r + 1 > d$ possible values. Polynomial interpolation says that every coefficient in that variable vanishes. Repeating this argument in the other variables proves $f = 0$, a contradiction. □

We use here the presentation and Schubert representatives of [4]. Localizing the equivariant coefficient ring does not affect the conclusion, since clearing its denominators is injective.

4.5 The line Chevalley formula

The proof of Theorem 2 is an induction on k driven by multiplication by the single line class

$$T_k = (1 - \beta^2 q_k) X_k = (1 - \beta^2 q_k)(1 + \beta x_k),$$

exactly as the proof of the quantum Schubert Pieri formula is driven by multiplication by $x_k - z$. The following lemma is the K-theoretic single-variable Monk formula.

Definition 13 (Line paths). *Let $k \geq 1$ and $u \in S_\infty$, and put $M = \max\{n(u), k\} + 1$ as in Proposition 5. A* line path at k from u *is a directed path π in the quantum Bruhat graph with labels*

$$(a_1, k),\ (a_2, k),\ \ldots,\ (a_s, k), \qquad (k, b_1),\ (k, b_2),\ \ldots,\ (k, b_t),$$

where $s, t \geq 0$, $k > a_1 > a_2 > \cdots > a_s \geq 1$, and $M \geq b_1 > b_2 > \cdots > b_t > k$. Write $s(\pi)$ and $t(\pi)$ for the two lengths, $v(\pi)$ for the permutation reached after the first $s(\pi)$ edges, $\mathrm{end}(\pi)$ *for the final permutation, and $D(\pi)$ for the quantum degree, defined edgewise as in Definition 1. The empty path is allowed, with $v(\pi) = \mathrm{end}(\pi) = u$.*

Lemma 14 (Line Chevalley formula). *For every $k \geq 1$ and $u \in S_\infty$,*

$$T_k \, \mathfrak{G}^q_u(x; y) = \sum_\pi (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \, \frac{1}{1 + \beta y_{v(\pi)(k)}} \, \mathfrak{G}^q_{\mathrm{end}(\pi)}(x; y), \tag{11}$$

the sum running over all line paths at k from u.

Proof. Set $\beta=-1$ first and write $Y_a=(1-y_a)^{-1}$, so that the claimed coefficient is $(-1)^{t(\pi)}q^{D(\pi)}Y_{v(\pi)(k)}$. Work in $QK_T(Fl_N)$ for N large, with the presentation and Schubert representatives of [4]: the residue class of T_k is the line bundle class $[\mathcal{O}(-\epsilon_k)]$ with $\epsilon_k=\varpi_k-\varpi_{k-1}$, $\mathfrak{G}^q_w$ represents the opposite Schubert class $[\mathcal{O}^w]$, and the equivariant parameters are identified by $1-y_a=\mathbf{e}^{-\epsilon_a}$.

Kato's isomorphism Φ onto the equivariant K-group of the semi-infinite flag manifold sends $\mathbf{e}^{\mu}[\mathcal{O}^w]Q^{\xi}$ to $\mathbf{e}^{-\mu}[\mathcal{O}_{\mathbf{Q}}(wt_\xi)]$, and by [4, Section 3.1], quoting [3, Proposition 5.2], it intertwines quantum multiplication by $[\mathcal{O}(-\epsilon_k)]$ with $(1-\mathbf{t}_{k-1})\,[\mathcal{O}_{\mathbf{Q}}(w_0\epsilon_k)]\otimes(-)$, where $\mathbf{t}_{k-1}$ corresponds to Q_{k-1} and $\mathbf{t}_0=0$. The tensor product with $[\mathcal{O}_{\mathbf{Q}}(w_0\epsilon_k)]$ is computed by the general Chevalley formula of Lenart–Naito–Sagaki [1, Theorem 33], in the form recalled in [4, Section 5.1], for the weight $\lambda=-\epsilon_k$ and any reduced λ-chain. We use the chain of [4, Section 5.2]: the alcove path from $A_\circ$ to A_{ϵ_k} crosses exactly the hyperplanes

$$H_{\epsilon_a-\epsilon_k,\,0}\ (a=k-1,\dots,1),\qquad H_{\epsilon_k-\epsilon_b,\,1}\ (b=N,\dots,k+1),$$

in this order. The first group is crossed in the negative direction, so the corresponding roots of the chain are the positive roots $\epsilon_a-\epsilon_k$; the second group is crossed in the positive direction, so the corresponding roots are the negative roots $-(\epsilon_k-\epsilon_b)$. An admissible subset is therefore a line path π, and the statistic $n(A)$ counting negative roots is $t(\pi)$. The affine reflections of the first group are linear, while those of the second group fix ϵ_k, because $\epsilon_k\in H_{\epsilon_k-\epsilon_b,1}$; hence the weight statistic is $-v(\pi)\epsilon_k=-\epsilon_{v(\pi)(k)}$. Finally, $\langle-\epsilon_k,\alpha^\vee_{k-1}\rangle=1$ and all other pairings of $-\epsilon_k$ with simple coroots are nonpositive, so the auxiliary sum over partitions in the general formula is a geometric series contributing $(1-\mathbf{t}_{k-1})^{-1}$, which cancels the factor $1-\mathbf{t}_{k-1}$ above. Transporting back through Φ, which negates the exponent of $\mathbf{e}$ and converts t_ξ into Q^ξ, gives

$$T_k\star[\mathcal{O}^u]=\sum_{\pi}(-1)^{t(\pi)}Q^{D(\pi)}\mathbf{e}^{\epsilon_{v(\pi)(k)}}[\mathcal{O}^{\mathrm{end}(\pi)}]=\sum_{\pi}(-1)^{t(\pi)}q^{D(\pi)}Y_{v(\pi)(k)}[\mathcal{O}^{\mathrm{end}(\pi)}]$$

in $QK_T(Fl_N)$. For $k=1$ there are no left edges and this is the first fundamental weight case of the Chevalley formula in [1, Section 5], after the substitution $[\mathcal{O}^{s_1}]=1-\mathbf{e}^{-\varpi_1}T_1$.

The bound $b_1\le M$ is proved as in Proposition 5: $v(\pi)$ agrees with u outside $\{1,\dots,k\}$, so if $b_1>M$ then $v(\pi)(b_1)=b_1$, $v(\pi)(b_1-1)=b_1-1$ and $v(\pi)(k)<b_1-1$, and (k,b_1) is neither a Bruhat nor a quantum edge. Thus the right side of (11) is independent of N, and Proposition 12 upgrades the identity in $QK_T(Fl_N)$ to a polynomial identity at $\beta=-1$. The homogenization to general β is the scaling used in Proposition 15 below: with $c=-\beta$ and $(x,y,q)\mapsto(cx,cy,\beta^2q)$ one has $T_k\mapsto T_k$, $Y_a\mapsto(1+\beta y_a)^{-1}$, $\mathfrak{G}^q_w\mapsto c^{\ell(w)}\mathfrak{G}^q_w$, and

$$c^{\ell(\mathrm{end}(\pi))-\ell(u)}\beta^{2|D(\pi)|}(-1)^{t(\pi)}=(-\beta)^{s(\pi)}\beta^{t(\pi)}$$

by the length–degree identity $\ell(\mathrm{end}(\pi))-\ell(u)=s(\pi)+t(\pi)-2|D(\pi)|$. □

We record the operator form in which the lemma is used. For a formal combination $\Theta=\sum_v\theta_v\,\mathfrak{G}^q_v$ whose coefficients are free of x, put

$$\mathcal{T}_k\Theta:=\sum_v\theta_v\sum_{\pi\text{ from }v}(-\beta)^{s(\pi)}\beta^{t(\pi)}q^{D(\pi)}\,\frac{\mathfrak{G}^q_{\mathrm{end}(\pi)}}{1+\beta y_{v(\pi)(k)}},\tag{12}$$

so that Lemma 14 reads $T_k\Theta=\mathcal{T}_k\Theta$ as polynomials.

4.6 Proof of the triple formula for $k = 1$

Proposition 15. *Theorem 2 holds for $k = 1$ and arbitrary $u \in S_\infty$.*

Proof. Take $k = 1$ in Lemma 14. There are no left edges, so $v(\pi) = u$ for every line path, the equivariant factor is the constant $(1 + \beta y_{u(1)})^{-1}$, and

$$(1 - \beta^2 q_1) X_1 \mathfrak{G}_u^q = \frac{1}{1 + \beta y_{u(1)}} \sum_\gamma \beta^{s(\gamma)} q^{D(\gamma)} \mathfrak{G}_{\mathrm{end}(\gamma)}^q, \tag{13}$$

where γ runs over directed paths with labels forming a subsequence of $(1, M), (1, M-1), \ldots, (1, 2)$. These are exactly the admissible chains of Definition 1 for $k = 1$: all lower indices are 1, and distinctness and weak decrease of the upper indices make them strictly decreasing. There is at most one such path to a given endpoint. In fact, a nonempty product $t_{1b_1} \cdots t_{1b_s}$ with distinct b_i is a single cycle on $\{1, b_1, \ldots, b_s\}$. The cycle of $u^{-1}w$ determines the ordered list of these indices. It also shows that a nonempty path cannot have endpoint u, and that the value $u(1)$ has moved out of position 1. Consequently a nonempty path has $m = 1$, Fix = Left = $\varnothing$, and Out = $\{u(1)\}$.

Finally, $\beta B_1^q = (1 - \beta^2 q_1) X_1 - (1 - \beta z)$. Subtract the second term from (13). The empty-path coefficient, after division by β, is

$$\frac{(1 + \beta y_{u(1)})^{-1} - (1 - \beta z)}{\beta} = z - \frac{y_{u(1)}}{1 + \beta y_{u(1)}}.$$

Every nonempty path has coefficient

$$q^{D(\gamma)} \frac{\beta^{s(\gamma)-1}}{1 + \beta y_{u(1)}}.$$

These are precisely (6) in the two cases. All exponents are nonnegative, so clearing equivariant denominators extends the result to $\beta = 0$ as well. □

For later use we make the scaling in the proof of Lemma 14 explicit. Put $c = -\beta$ and replace (x, y, z, q) at $\beta = -1$ by $(cx, cy, cz, \beta^2 q)$. The defining divided differences and the quantization map give

$$\mathfrak{G}_w^{\beta^2 q}(cx; cy)\big|_{\beta=-1} = c^{\ell(w)} \mathfrak{G}_w^q(x; y), \tag{14}$$

where on the left the quantum arguments are $\beta^2 q$ and the Grothendieck parameter is fixed at -1. The scaling takes place with β inverted; whenever both sides of a resulting identity have no negative powers of β, it holds before that inversion as well.

4.7 What the published marked-chain formula proves for all k

There is also an unconditional all-k calculation, but it is non-equivariant and initially indexed by paths rather than endpoints. For a nonempty admissible chain γ, let $r(\gamma)$ be the number of distinct lower indices a_i. Let $F(\gamma)$ be the following set of edge indices:

- every i in the maximal initial segment with $b_1 = \cdots = b_i$ and $a_1 > \cdots > a_i$;
- every $i < s(\gamma)$ for which $(a_i, b_i) \succ (a_{i+1}, b_{i+1})$.

Put $h(\gamma) = |F(\gamma)|$. These are exactly the indices forced to be marked in the Naito–Sagaki marking rule; in particular $h(\gamma) \geq 1$.

Proposition 16. *For every $k \geq 1$ and $u \in S_\infty$,*

$$B_k^q(x;z)\mathfrak{G}_u^q(x;0) = z^k \mathfrak{G}_u^q(x;0) + \sum_{\substack{\gamma \text{ admissible} \\ s(\gamma)>0}} q^{D(\gamma)} \beta^{s(\gamma)-r(\gamma)} z^{k-r(\gamma)} (1-\beta z)^{h(\gamma)-1} \mathfrak{G}^q_{\mathrm{end}(\gamma)}(x;0). \tag{15}$$

This statement does not require uniqueness of a path to its endpoint.

Proof. For $1 \leq p \leq k$, let $c[k,p] = (k-p+1, k-p+2, \ldots, k+1)$, in cycle notation. The single Grothendieck polynomial of this cycle is

$$\mathfrak{G}_{c[k,p]}(x;0) = \sum_{j=p}^{k} \binom{j-1}{p-1} \beta^{j-p} e_j(x_1, \ldots, x_k).$$

The binomial theorem, coefficient by coefficient in e_j, gives

$$\prod_{i=1}^{k}(x_i + z) = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1} \mathfrak{G}_{c[k,p]}(x;0).$$

Indeed the coefficient of e_j on the right is $z^{k-j}\big((1-\beta z)+\beta z\big)^{j-1} = z^{k-j}$. Applying the *linear* map $\mathcal{Q}$ gives

$$B_k^q = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1} \mathfrak{G}^q_{c[k,p]}(x;0). \tag{16}$$

No multiplicativity of $\mathcal{Q}$ is used.

Naito–Sagaki's Pieri theorem [5, Theorem 2.13], homogenized from $\beta = -1$, says that the contribution of γ to $\mathfrak{G}_u^q \mathfrak{G}^q_{c[k,p]}$ is

$$q^{D(\gamma)} \beta^{s(\gamma)-p} \# \operatorname{Mark}_p(\gamma).$$

A mark may occur only at a first occurrence of a lower index. All indices in $F(\gamma)$ must be marked; the remaining first occurrences can be chosen freely. The admissibility conditions ensure that forced indices are first occurrences. Thus, as also recorded in [5, Remark 2.14],

$$\# \operatorname{Mark}_p(\gamma) = \binom{r(\gamma)-h(\gamma)}{p-h(\gamma)}.$$

Here $1 \leq h(\gamma) \leq r(\gamma) \leq k$.

Fix a nonempty chain, and abbreviate its statistics by s, r, h, D. Its coefficient after multiplying (16) by $\mathfrak{G}_u^q$ is therefore

$$\begin{aligned} & q^D \sum_{p=h}^{r} \binom{r-h}{p-h} \beta^{s-p} z^{k-p} (1-\beta z)^{p-1} \\ &= q^D \beta^{s-r} z^{k-r} (1-\beta z)^{h-1} \sum_{p=h}^{r} \binom{r-h}{p-h} (\beta z)^{r-p} (1-\beta z)^{p-h} \\ &= q^D \beta^{s-r} z^{k-r} (1-\beta z)^{h-1}. \end{aligned}$$

The last sum is 1 by the binomial theorem. The separate term $z^k \mathfrak{G}_u^q$ supplies the empty path. This proves (15). Homogenization is justified by the same scaling and length–degree identity used in Proposition 15; the exponents $s-p$ are nonnegative. □

With Corollary 9 in hand, the path-indexed sum (15) is in fact indexed by endpoints, and its exponents can be read off from the endpoint. Write γ_w for the unique admissible k-chain from u to $w \in E(u,k)$.

Lemma 17 (Forced marks count left moves)**.** *Let $w \in E(u,k)$, $w \neq u$. Then*

$$r(\gamma_w) = m(u,w;k) \qquad \textit{and} \qquad h(\gamma_w) = |\operatorname{Left}(u,w;k)| + 1.$$

Proof. The first equality is Corollary 9(2). For the second, note first that $(a_i,b_i) \succ (a_{i+1},b_{i+1})$ means $b_i = b_{i+1}$ and $a_i > a_{i+1}$; call such an index i a *descent* of γ_w. If i_0 is the last index of the maximal initial segment with equal upper indices and decreasing lower indices, then $1,\dots,i_0-1$ are descents and i_0 is not, so $F(\gamma_w)$ is the disjoint union of $\{i_0\}$ and the set of descents, and $h(\gamma_w)-1$ is the number of descents. We show that the descents are in bijection with $\operatorname{Left}(u,w;k)$.

By Corollary 9 we may rotate the values and assume that γ_w is a saturated chain in the k-Bruhat order satisfying (P0) and (P1); the sets $\operatorname{Fix}, \operatorname{Left}, \operatorname{Out}$ and the descents only involve positions and are unchanged. Consider a maximal run $(a_1,b),\dots,(a_p,b)$ of labels with the same upper index b, and let α_j be the value at position a_j just before the label (a_j,b) is applied, α_0 the value at b before the run. Each label is a Bruhat cover, so $\alpha_0 > \alpha_1 > \cdots > \alpha_p$, and the net effect of the run is that α_0 moves to a_1, α_j moves from a_j to a_{j+1} for $1 \le j < p$, and α_p moves from a_p to b. Position b is touched by no other label, so α_p stays at b for ever, and every value entering the window from outside does so at the start of a run. In particular a value moves within the window only in steps $a_j \to a_{j+1}$, which is a move to the left exactly at a descent.

Let i be a descent, so that the labels i and $i+1$ are consecutive labels $(a_j,b),(a_{j+1},b)$ of a run with $a_{j+1} < a_j$, and the value α_j moves from a_j to a_{j+1}. By (P0) the lower index a_j is a first occurrence, so position a_j was untouched and $\alpha_j = u(a_j)$ is a window value making its first move. By the prohibition lemma of Lenart and Sottile [2, Lemma 2.9], a chain satisfying (P0) and (P1) contains no subsequence $(j',m),\dots,(i',m),\dots,(i',l)$ with $i' < j' \le k < l < m$; applied with $(j',m) = (a_j,b)$ and $(i',m) = (a_{j+1},b)$, this says that position a_{j+1} is never touched again (a later label (a_{j+1},l) has $l < b$ by (P1) and distinctness). Hence $u(a_j)$ ends at position $a_{j+1} < a_j$ and lies in $\operatorname{Left}(u,w;k)$. Distinct descents move distinct values, so this gives an injection from descents into $\operatorname{Left}(u,w;k)$.

Conversely let $c = u(i) \in \operatorname{Left}(u,w;k)$. It moves at least once; its first move is by a label (i,b). If this label ends its run, c stays at b and is in Out; otherwise c moves to the lower index a' of the next label. If $a' < i$ this is a descent, and c is the value attached to it. If $a' > i$, then every later move of c starts from a position that has already been touched, so by (P0) it is a step to a larger lower index or the end of a run; thus c only moves right or leaves the window, and ends in Out, a contradiction. Hence every value in $\operatorname{Left}(u,w;k)$ is attached to a descent, and the correspondence is a bijection. □

Theorem 18 (The non-equivariant case)**.** *Theorem 2 holds at $y = 0$: for every $k \ge 1$ and $u \in S_\infty$,*

$$B_k^q(x;z)\,\mathfrak{G}_u^q(x;0) = \sum_{w \in E(u,k)} q^{D(u,w;k)}\beta^{s(u,w;k)-m(u,w;k)}z^{|\operatorname{Fix}(u,w;k)|}(1-\beta z)^{|\operatorname{Left}(u,w;k)|}\,\mathfrak{G}_w^q(x;0).$$

Proof. By Corollary 9(1) the sum in (15) has exactly one term for each $w \in E(u,k)\setminus\{u\}$, namely $\gamma = \gamma_w$, and the term $z^k\mathfrak{G}_u^q$ is the term $w = u$ with $\operatorname{Fix} = \{u(1),\dots,u(k)\}$. By Lemma 17, $k - r(\gamma_w) = k - m = |\operatorname{Fix}|$, $s(\gamma_w) - r(\gamma_w) = s - m$ and $h(\gamma_w) - 1 = |\operatorname{Left}|$. This is (6) at $y = 0$. □

Thus the only part of Theorem 2 that remains open is the dependence on the equivariant parameters y. At $z = 1/\beta$ this dependence is settled in Theorem 22 below.

4.8 Reduction of level k to levels $k-1$ and $k-2$

We now set up the induction on k. Throughout, $a = 1-\beta z$ and H_j denotes the tiling polynomial introduced in the proof of Proposition 11:

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z)H_{j-1} + a\, q_{j-1}X_{j-1}H_{j-2} \quad (j \geq 2),$$

and $B^q_j = H_j - \beta q_j X_j H_{j-1}$ for $j \geq 1$ by (10). We use the convention $q_0 = 0$ and $B^q_{-1} = 0$.

Lemma 19 (Two-step block identities)**.** *For every $k \geq 1$,*

$$\beta B^q_k + aB^q_{k-1} = T_k H_{k-1}, \tag{17}$$

$$(1-\beta^2 q_{k-1})H_{k-1} = B^q_{k-1} + a\beta q_{k-1}B^q_{k-2}. \tag{18}$$

Consequently

$$(1-\beta^2 q_{k-1})\big(\beta B^q_k + aB^q_{k-1}\big) = T_k\big(B^q_{k-1} + a\beta q_{k-1}B^q_{k-2}\big). \tag{19}$$

Proof. Substituting $B^q_k = H_k - \beta q_k X_k H_{k-1}$, $B^q_{k-1} = H_{k-1} - \beta q_{k-1}X_{k-1}H_{k-2}$ and the recurrence for H_k,

$$\beta B^q_k + aB^q_{k-1} = \big(\beta(x_k+z) + a - \beta^2 q_k X_k\big)H_{k-1} = (1-\beta^2 q_k)X_k H_{k-1},$$

since $\beta(x_k+z)+a = 1+\beta x_k = X_k$; the two terms involving $a\beta q_{k-1}X_{k-1}H_{k-2}$ cancel. This is (17). For (18) with $k \geq 3$, expand both sides using the recurrence for H_{k-1} and the expressions of B^q_{k-1}, B^q_{k-2} through H: the right side is

$$H_{k-1} + \beta q_{k-1}(a - X_{k-1})H_{k-2} - a\beta^2 q_{k-1}q_{k-2}X_{k-2}H_{k-3},$$

and $\beta(a - X_{k-1}) = -\beta^2(x_{k-1}+z)$, so this equals $H_{k-1} - \beta^2 q_{k-1}\big((x_{k-1}+z)H_{k-2} + aq_{k-2}X_{k-2}H_{k-3}\big) = (1-\beta^2 q_{k-1})H_{k-1}$. For $k=2$ the same computation applies without the last term, and for $k=1$ both sides equal 1. Multiplying (17) by $1-\beta^2 q_{k-1}$ and substituting (18) gives (19). □

At $\beta = 0$, (19) reduces to $B_k = (x_k+z)B_{k-1} + q_{k-1}B_{k-2}$, the recurrence used for the quantum Schubert Pieri formula. At $\beta = -1$ it reads $(1-q_{k-1})\big(AB_{k-1} - B_k\big) = T_k\big(B_{k-1} - Aq_{k-1}B_{k-2}\big)$ with $A = 1+z$.

The target identity. For $j \geq 0$ and $u \in S_\infty$ write

$$P_j(u) := \sum_{w \in E(u,j)} C^{(j)}_{u,w}\, \mathfrak{G}^q_w, \qquad P_0(u) = \mathfrak{G}^q_u, \qquad P_{-1}(u) = 0,$$

for the right side of (5) at level j, with $C^{(j)}_{u,w}$ given by (6). The endpoint data (s, D) entering $C^{(j)}_{u,w}$ are well defined by Corollary 9.

Proposition 20 (Reduction)**.** *Let $k \geq 1$ and suppose Theorem 2 holds at levels $k-1$ and $k-2$ for the permutation u, that is, $B^q_{k-1}\mathfrak{G}^q_u = P_{k-1}(u)$ and $B^q_{k-2}\mathfrak{G}^q_u = P_{k-2}(u)$. Then Theorem 2 holds at level k for u if and only if*

$$\beta(1-\beta^2 q_{k-1})\, P_k(u) = \mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big) - a(1-\beta^2 q_{k-1})\, P_{k-1}(u), \tag{20}$$

with $\mathcal{T}_k$ the line operator (12)*.*

Proof. By (19) and the hypotheses,

$$(1-\beta^2 q_{k-1})\big(\beta B_k^q \mathfrak{G}_u^q + aP_{k-1}(u)\big) = T_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big),$$

and the right side equals $\mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)\big)$ by Lemma 14. Hence $\beta(1-\beta^2 q_{k-1})B_k^q\mathfrak{G}_u^q$ equals the right side of (20). Since the $\mathfrak{G}_w^q$ are linearly independent over the coefficient field and $\beta(1-\beta^2 q_{k-1})$ is a nonzero element of the integral domain of coefficients, $B_k^q\mathfrak{G}_u^q = P_k(u)$ holds if and only if (20) does. □

For $k=1$ the identity (20) is $\beta P_1(u) = \mathcal{T}_1\mathfrak{G}_u^q - a\mathfrak{G}_u^q$, which is exactly the computation in the proof of Proposition 15. Thus Theorem 2 follows, by induction on k, from (20) for all $k \geq 2$ and all u. We call (20) the *junction identity*; it is the K-theoretic counterpart of the recurrence $\mathcal{E}_{k,k} = (x_k - z_1)\mathcal{E}_{k-1,k-1} + q_{k-1}\mathcal{E}_{k-2,k-2}$ for quantum factorial elementary polynomials expanded by the single-variable Monk formula in the quantum Schubert case.

Coefficient form. Extracting the coefficient of $\mathfrak{G}_w^q$ in (20), and writing $\pi\colon v \to w$ for a line path at k from v with endpoint w, the junction identity reads

$$\begin{aligned}\beta(1-\beta^2 q_{k-1})\,C_{u,w}^{(k)} = \sum_v \sum_{\pi\colon v\to w} \frac{(-\beta)^{s(\pi)}\beta^{t(\pi)}q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}}\Big(C_{u,v}^{(k-1)} + a\beta q_{k-1}C_{u,v}^{(k-2)}\Big)\\ - a(1-\beta^2 q_{k-1})\,C_{u,w}^{(k-1)},\end{aligned} \tag{21}$$

where $C_{u,v}^{(j)} = 0$ for $v \notin E(u,j)$ and $C^{(-1)} = 0$. A line path $\pi\colon v \to w$ is determined by its label sequence, and $v = w\,t_{kb_t}\cdots t_{kb_1}t_{a_sk}\cdots t_{a_1k}$; the path must be directed from v in the quantum Bruhat graph. The term $\pi = \varnothing$, $v = w$, is the diagonal term, with equivariant factor $(1+\beta y_{w(k)})^{-1}$; the paths with $s(\pi) = 0$ are the up-neighbours $v = wt_{kb_t}\cdots t_{kb_1}$, and the paths with $t(\pi) = 0$ are the down-neighbours $v = wt_{a_sk}\cdots t_{a_1k}$. In the quantum Schubert case only single edges occur and (21) becomes the four-term expansion of that proof; here a path may combine several down edges followed by several up edges, and each term carries a K-theoretic sign and an equivariant denominator determined by the intermediate permutation $v(\pi)$.

Remark 21 (Well-definedness)**.** *The identity* (21) *is to be proved for each endpoint w. By Corollary 9 the data (s, D) entering $C_{u,w}^{(k)}$ are already known to be well defined, and D is given by* (7)*; so the left side is unambiguous, and a proof of* (21) *need not establish path-independence separately. It does, however, have to show that the right side collapses to the single monomial $q^{D(u,w;k)}$ times the expected endpoint factors, and that it vanishes for $w \notin E(u,k)$. In the quantum Schubert proof the analogous path-independence was proved through the cycle structure of $u^{-1}w$; here it comes from the rotation to the classical k-Bruhat order.*

The diagonal case. Suppose $u(k) = w(k)$. Position k is touched by a k-chain only through labels (k,b), and by Corollary 9(2) a touched position never regains its original value. Hence the admissible k-chains from u to w have no label with lower index k; they are exactly the admissible $(k-1)$-chains from u to w all of whose labels have second entry at least $k+1$, and for these chains the conditions of Definition 1 at levels k and $k-1$ coincide. Corollary 9(2) at level $k-1$, where k is an outside position, shows likewise that the $(k-1)$-chains to w with some label (a,k) have $w(k) \neq u(k)$. Therefore

$$\mathrm{Fix}(u,w;k) = \mathrm{Fix}(u,w;k-1) \sqcup \{u(k)\}, \qquad \text{Left, Out, } s, \ D \text{ unchanged},$$

and

$$C^{(k)}_{u,w} = \Big(z - \frac{y_{u(k)}}{1+\beta y_{u(k)}}\Big) C^{(k-1)}_{u,w} = \frac{(1+\beta y_{u(k)})^{-1} - a}{\beta}\, C^{(k-1)}_{u,w}. \tag{22}$$

Inserting (22) into (21), the diagonal term $\pi = \varnothing$ together with the last term produce exactly $\beta C^{(k)}_{u,w}$ plus $\frac{a\beta q_{k-1}}{1+\beta y_{u(k)}} C^{(k-2)}_{u,w}$, and what remains to be shown in the case $u(k) = w(k)$ is the cancellation

$$\sum_{v} \sum_{\substack{\pi:\, v \to w \\ \pi \neq \varnothing}} \frac{(-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}} \Big(C^{(k-1)}_{u,v} + a\beta q_{k-1} C^{(k-2)}_{u,v} \Big) + \frac{\beta q_{k-1}}{1+\beta y_{u(k)}} \Big(\beta C^{(k-1)}_{u,w} + a C^{(k-2)}_{u,w} \Big) = 0, \tag{23}$$

where we used $\beta(z+\overline{y}_{u(k)}) + a = (1+\beta y_{u(k)})^{-1}$ to combine the two multiples of $q_{k-1} C^{(k-1)}_{u,w}$. This is the K-theoretic form of the reflection–drop cancellation of the quantum Schubert proof: there the nonempty paths are single edges t_{ak}, t_{kb} and the drop term is $q_{k-1} c_{k-2}(u,w)$. Note the combination $\beta C^{(k-1)}_{u,w} + a C^{(k-2)}_{u,w}$, the coefficient analogue of $\beta B^q_{k-1} + a B^q_{k-2} = T_{k-1} H_{k-2}$.

4.9 The case $z = 1/\beta$: the Chevalley formula for ϖ_k

At $z = 1/\beta$, that is $a = 0$, the junction identity loses its level-$(k-2)$ part and its last term, and we can prove it completely. This gives the full equivariant statement of Theorem 2 on the hyperplane $a = 0$, for all k and u. Since $x_i + 1/\beta = X_i/\beta$, the block is $B^q_k|_{z=1/\beta} = \beta^{-k} \mathcal{Q}(X_1 \cdots X_k) = \beta^{-k} F^k_k = \beta^{-k}(1-\beta^2 q_k) X_1 \cdots X_k$, and the coefficient (6) specializes by $(1-\beta z)^{|\,\mathrm{Left}\,|} = 0^{|\,\mathrm{Left}\,|}$ and $z + \overline{y}_c = (1+\beta y_c)^{-1}/\beta$ to

$$C^{(k)}_{u,w}\big|_{z=1/\beta} = \begin{cases} q^D \beta^{s-k} \prod_{i=1}^{k} (1+\beta y_{u(i)})^{-1} & \mathrm{Left}(u,w;k) = \varnothing, \\ 0 & \text{otherwise}, \end{cases}$$

using $m + |\,\mathrm{Fix}\,| = k$ and $\mathrm{Fix} \sqcup \mathrm{Out} = \{u(1), \ldots, u(k)\}$ when $\mathrm{Left} = \varnothing$. Thus the statement to be proved is the following.

Theorem 22 (Theorem 2 at $z = 1/\beta$)**.** *For every $k \geq 1$ and $u \in S_\infty$,*

$$(1-\beta^2 q_k)\, X_1 X_2 \cdots X_k\, \mathfrak{G}^q_u(x;y) = \prod_{i=1}^{k} \frac{1}{1+\beta y_{u(i)}} \sum_{\substack{w \in E(u,k) \\ \mathrm{Left}(u,w;k) = \varnothing}} \beta^{s(u,w;k)} q^{D(u,w;k)}\, \mathfrak{G}^q_w(x;y). \tag{24}$$

The polynomial $(1-\beta^2 q_k) X_1 \cdots X_k = F^k_k$ represents, in $QK_T(Fl_N)$ with the identifications used in the proof of Lemma 14, the class of the line bundle $\mathcal{O}(-\varpi_k)$: by [4, Section 3.1], $[\mathcal{O}(-\varpi_k)]$ is the quantum product of $[\mathcal{O}(-\epsilon_1)]$, $\frac{1}{1-Q_1}[\mathcal{O}(-\epsilon_2)], \ldots, \frac{1}{1-Q_{k-1}}[\mathcal{O}(-\epsilon_k)]$, and $\frac{1}{1-Q_{j-1}}[\mathcal{O}(-\epsilon_j)]$ is represented by $\frac{1-\beta^2 q_j}{1-\beta^2 q_{j-1}} X_j$. So (24) is a chain-indexed form of the equivariant quantum K-Chevalley formula for $\mathcal{O}(-\varpi_k)$, with the equivariant weight $\prod_{i \leq k} (1+\beta y_{u(i)})^{-1}$ independent of the endpoint. We prove it from the single-variable formula (Lemma 14) by the junction identity, and we record the combinatorial lemmas separately, since they are also the ingredients of the general case.

Left-free chains. Call an admissible k-chain, or its endpoint, *Left-free* if $\mathrm{Left}(u,w;k) = \varnothing$. Recall that $\prec$ is the total order on labels with $(a,b) \prec (c,d)$ if $b > d$, or $b = d$ and $a < c$.

Lemma 23 (Left-free chains)**.** *An admissible k-chain is Left-free if and only if its label sequence is strictly increasing for $\prec$. Conversely, every directed path in the quantum Bruhat graph from u whose labels (a_i, b_i) satisfy $a_i \le k < b_i$ and are strictly increasing for $\prec$ is an admissible k-chain, and it is Left-free.*

Proof. For an admissible chain the upper indices are weakly decreasing, so the sequence is strictly $\prec$-increasing exactly when there is no index i with $b_i = b_{i+1}$ and $a_i > a_{i+1}$, that is, no descent in the sense of the proof of Lemma 17. That proof shows that the descents are in bijection with $\mathrm{Left}(u, w; k)$. For the converse, a strictly $\prec$-increasing sequence has distinct labels and weakly decreasing upper indices, and condition 4 of Definition 1 holds for every index, repeated lower index or not. Hence the path is an admissible chain, and it is Left-free by the first part. □

Left-free chains are therefore the paths of the quantum alcove model for the reflection order $\prec$ on the roots $\epsilon_a - \epsilon_b$, $a \le k < b$, which is consistent with the Chevalley interpretation of (24).

Three lemmas on rotated paths. Fix $u \in S_R$ and k, and let $\vartheta = u(k+1)$ and $\rho = \rho_\vartheta$ be as in Lemma 8. For $v \in S_R$ and $1 \le j < R$ put $\sigma_j(v) = \#\{i \le j : \ v(i) \le \vartheta\}$.

Lemma 24 (Degree along rotated paths)**.** *Let $u = v_0 \to v_1 \to \cdots \to v_s = w$ be a directed path in the quantum Bruhat graph with labels (a_i, b_i), $a_i \le k < b_i$, such that every edge of the rotated path $\rho v_0 \to \cdots \to \rho v_s$ is a Bruhat edge. Then $D_j = \sigma_j(w) - \sigma_j(u)$ for every j, where D_j is the number of quantum edges of the path with $a_i \le j < b_i$.*

Proof. An edge (a, b) with $a \le j < b$ applied to v changes σ_j by $+1$ if $v(b) \le \vartheta < v(a)$, by -1 if $v(a) \le \vartheta < v(b)$, and not at all otherwise. By Corollary 7, since the rotated edge is Bruhat, the second case does not occur, and the first case occurs exactly when the edge is quantum. Edges with both positions at most j or both greater than j do not change σ_j and are not counted by D_j. Summing along the path gives the claim. □

Lemma 25 (Commuting a right edge)**.** *Let $x \in S_R$ and $a < k < b' < b$. Then $x \lessdot xt_{kb} \lessdot xt_{kb}t_{ab'}$ is a chain of Bruhat covers if and only if $x \lessdot xt_{ab'} \lessdot xt_{ab'}t_{kb}$ is. The two chains have the same endpoint.*

Proof. The transpositions commute, so the endpoints agree. Write $A = x(a)$, $K = x(k)$, $B' = x(b')$ and $B = x(b)$. Recall that $y \lessdot yt_{ij}$ means $y(i) < y(j)$ and that no value at a position strictly between i and j lies in the interval $(y(i), y(j))$. The positions strictly between k and b other than b', and the positions strictly between a and b' other than k, carry the same values in all four permutations involved, so only the four values A, K, B', B matter. The first chain is a chain of covers if and only if

$$K < B, \quad B' \notin (K, B), \quad A < B', \quad B \notin (A, B')$$

together with the conditions on the other positions, and the second if and only if

$$A < B', \quad K \notin (A, B'), \quad K < B, \quad A \notin (K, B)$$

together with the same conditions on the other positions. Assume the first set. If $B' < K$, then $K \notin (A, B')$ and $A < B' < K$ give $A \notin (K, B)$. If $B' > B$, then $B \in (K, B')$ and $B \notin (A, B')$ force $A > B > K$, so again $K \notin (A, B')$ and $A \notin (K, B)$. Conversely assume the second set. If $K < A$, then $A \notin (K, B)$ forces $A > B$, so $B < A < B'$, whence $B \notin (A, B')$ and $B' \notin (K, B)$. If $K > B'$, then $B' < K$ gives $B' \notin (K, B)$ and $B > K > B'$ gives $B \notin (A, B')$. □

For a sequence γ of labels (a, b) with $a \leq k < b$ let $\mathrm{sort}(\gamma)$ be the sequence obtained by moving all labels with lower index k to the end, preserving the relative order of these labels and of the remaining labels.

Lemma 26 (Sorting)**.** *Let γ be a strictly $\prec$-increasing sequence of labels (a, b) with $a \leq k < b$, and let $x \in S_R$. Then γ is a chain of Bruhat covers from x if and only if* $\mathrm{sort}(\gamma)$ *is, and the two chains have the same endpoint.*

Proof. If a label (k, b) precedes a label (a, b') with $a < k$ in γ, then $(k, b) \prec (a, b')$ forces $b' < b$, since $b' = b$ would require $k < a$. The sorted sequence is obtained from γ by repeatedly interchanging an adjacent pair $(k, b), (a, b')$ with $a < k$ into $(a, b'), (k, b)$; the pairs interchanged are always in their original relative order, so $b' < b$ holds at each step. Each step preserves the endpoint and, by Lemma 25 applied to the permutation reached before the pair, the property of being a chain of covers. □

Lemma 27 (Rotation of sorted words)**.** *Let $\omega = \gamma'\,(k, b_1) \cdots (k, b_r)$ be a directed path in the quantum Bruhat graph from u, where the labels of γ' have lower index at most $k-1$ and upper index at least $k+1$, are pairwise distinct with weakly decreasing upper indices, and $b_1 > \cdots > b_r > k$. Then every edge of the rotated path $\rho u \to \cdots$ is a Bruhat edge.*

Proof. The labels of γ' satisfy the hypotheses of Lemma 8, so the rotated edges of γ' are Bruhat. Work in the rotated order, with $N = \rho(\vartheta)$ the largest value, initially at position $k+1$. Let $(c_1, k+1), \ldots, (c_m, k+1)$ be the labels of γ' with upper index $k+1$; they are its last m labels. As in the proof of Lemma 8, put $e_0 = N$ and let e_h be the value at position c_h just before $(c_h, k+1)$; then $e_0 > e_1 > \cdots > e_m$, after γ' the value at position $k+1$ is e_m, and e_{h-1} sits at position c_h. Position k is touched by no label of γ'; let x be its value. Since $c_h < k < k+1$, the Bruhat condition for $(c_h, k+1)$ gives $x \notin (e_h, e_{h-1})$ for each h, and x differs from all e_h. Hence $x \notin (e_m, N)$, and as $x < N$ we get $x < e_m$.

Let $y_0 = x$ and let y_i be the value at position k after the edge (k, b_i). We show by induction on i that (k, b_i) is Bruhat and that $e_m > y_i$ whenever $b_i \neq k+1$; the induction hypothesis is $e_m > y_{i-1}$, true for $i = 1$. Position $k+1$ still carries e_m when (k, b_i) is applied, because $b_1, \ldots, b_{i-1} > b_i \geq k+1$. If (k, b_i) were quantum, every value at a position strictly between k and b_i would lie in $(v(b_i), y_{i-1})$, and $v(b_i) < y_{i-1}$; for $b_i > k+1$ this puts e_m below y_{i-1}, and for $b_i = k+1$ it says $e_m = v(b_i) < y_{i-1}$; both contradict $e_m > y_{i-1}$. So the edge is Bruhat, $y_i = v(b_i) > y_{i-1}$, and if $b_i > k+1$ the Bruhat condition gives $e_m \notin (y_{i-1}, y_i)$, whence $e_m > y_i$. If $b_i = k+1$ it is the last right edge. □

Corollary 28 (Sorting bijection)**.** *Let $w \in S_R$. The map $\gamma \mapsto \mathrm{sort}(\gamma)$ is a bijection from the Left-free admissible k-chains from u to w onto the directed paths from u to w of the form $\omega = \gamma'\,(k, b_1) \cdots (k, b_r)$, where γ' is a Left-free admissible $(k-1)$-chain from u none of whose labels has upper index k, and $b_1 > \cdots > b_r > k$. It preserves the number of edges and the quantum degree D.*

Proof. Let γ be a Left-free k-chain from u to w. By Lemma 8 the rotated path is a chain of Bruhat covers from ρu to ρw; by Lemma 23 its labels are strictly $\prec$-increasing, so by Lemma 26 $\mathrm{sort}(\gamma)$ is a chain of Bruhat covers from ρu to ρw, and by Corollary 7 it is a directed path from u to w in the quantum Bruhat graph. Its labels with lower index less than k form a strictly $\prec$-increasing directed path γ' from u with upper indices greater than k, hence a Left-free $(k-1)$-chain by Lemma 23, and its labels (k, b) appear in $\prec$-order, that is with decreasing b. Both γ and $\mathrm{sort}(\gamma)$ have all rotated edges Bruhat, so by Lemma 24 they have the same D, and they have the same number of edges.

Conversely let ω be a path of the stated form. By Lemma 27 its rotated edges are Bruhat. Let γ be the $\prec$-increasing rearrangement of its labels, so that $\mathrm{sort}(\gamma) = \omega$; the labels are distinct since

γ' has distinct labels and the (k, b_i) are distinct from them. By Lemma 26 γ is a chain of Bruhat covers from ρu to ρw, hence a directed path from u to w, and by Lemma 23 a Left-free k-chain. The two constructions are inverse to each other. $\square$

Proof of Theorem 22. We induct on k, reading the right side of (24) as $\mathfrak{G}_u^q$ for $k = 0$, where the statement is trivial. Let $k \geq 1$ and assume (24) at level $k-1$. Since $(1-\beta^2 q_{k-1})F_k^k = T_k F_{k-1}^{k-1}$, the induction hypothesis and Lemma 14 give

$$(1-\beta^2 q_{k-1})F_k^k\, \mathfrak{G}_u^q = \prod_{i=1}^{k-1} \frac{1}{1+\beta y_{u(i)}} \sum_{\gamma} \beta^{s(\gamma)} q^{D(\gamma)}\, \mathcal{T}_k\, \mathfrak{G}_{\mathrm{end}(\gamma)}^q,$$

the sum over Left-free $(k-1)$-chains γ from u (one for each Left-free endpoint, by Corollary 9). Expanding $\mathcal{T}_k$ by (12), the coefficient of $\mathfrak{G}_w^q$ on the right is a sum over *composites* (γ, π), with π a line path at k from $\mathrm{end}(\gamma)$ to w, of the weights

$$\beta^{s(\gamma)} q^{D(\gamma)}\, (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)}\, \frac{1}{1+\beta y_{v(\pi)(k)}}.$$

Since the $\mathfrak{G}_w^q$ are linearly independent and $1-\beta^2 q_{k-1}$ is a nonzero polynomial, (24) at level k is equivalent to the identity, for every w,

$$\begin{aligned} \sum_{(\gamma,\pi)\colon\, u\to w} & \beta^{s(\gamma)} q^{D(\gamma)} (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \frac{1}{1+\beta y_{v(\pi)(k)}} \\ & = \frac{1-\beta^2 q_{k-1}}{1+\beta y_{u(k)}}\, \beta^{s(u,w;k)} q^{D(u,w;k)}\, [\, w \in E(u,k) \text{ Left-free}\,], \end{aligned} \tag{25}$$

which is the junction identity (21) at $a = 0$. We prove (25) by grouping composites by their underlying path.

Words and cuts. A composite (γ, π) determines the concatenated label sequence $\omega = \gamma\pi$, which we call its *word*; it is a directed path from u to w. By Lemma 23 the labels of γ are $\prec$-increasing, so its labels with upper index k, which have the largest possible $\prec$-value, form a terminal segment of γ with increasing lower indices; the left edges of π have upper index k and decreasing lower indices. Hence

$$\omega = \gamma_{>k}\ (a_1', k)\, (a_2', k) \cdots (a_m', k)\ (k, b_1) \cdots (k, b_r),$$

where $\gamma_{>k}$ consists of the labels of γ with upper index greater than k, the middle part is the *block* of ω, and $b_1 > \cdots > b_r > k$. The composite is recovered from ω by the *cut* $j \in \{0, \ldots, m\}$, the number of block labels belonging to γ. Conversely, given a directed path ω of this shape from u to w, the cut j yields a composite exactly when $\gamma_{>k}(a_1', k) \cdots (a_j', k)$ is a Left-free $(k-1)$-chain and $(a_{j+1}', k) \cdots (a_m', k)(k, b_1) \cdots (k, b_r)$ is a line path, that is, by Lemma 23 and Definition 13, exactly when

$$a_1' < a_2' < \cdots < a_j' \qquad \text{and} \qquad a_{j+1}' > a_{j+2}' > \cdots > a_m'.$$

All composites with word ω have the same endpoint, the same quantum degree $D(\omega)$, the same total number $|\omega|$ of labels, and the same value $v(\pi)(k)$, namely the value $c(\omega)$ at position k after the block, since position k is touched only by the block and the right edges. The weight of the composite with cut j is therefore

$$(-1)^{m-j}\, \beta^{|\omega|} q^{D(\omega)}\, \frac{1}{1+\beta y_{c(\omega)}}.$$

Words with two valid cuts cancel. Let $m \geq 1$. If j is a valid cut with $1 \leq j < m$, the largest block entry is a'_j or a'_{j+1}; if j and $j' > j$ are both valid, then $a'_{j+1}, \ldots, a'_{j'}$ lie both in an increasing prefix and in a decreasing suffix, so $j' = j+1$. Thus a word has at most two valid cuts, and if it has two, they are consecutive and their weights cancel. If j is valid, then $j+1$ is valid exactly when $a'_j < a'_{j+1}$, and $j-1$ is valid exactly when $a'_j > a'_{j+1}$ (for $1 \leq j < m$; for $j = 0$ or $j = m$ the block is monotone and the cut 1, respectively $m-1$, is valid as well). Hence a word with exactly one valid cut j has $1 \leq j < m$ and $a'_j = a'_{j+1}$: the same label (a'_j, k) is traversed twice in succession. Two consecutive traversals of a label (a, k) form a Bruhat edge followed by a quantum edge or vice versa, which forces $1 = 2(k-a)-1$, that is $a = k-1$. So a word with a single valid cut has block

$$a'_1 < \cdots < a'_{j-1} < k-1, \qquad a'_j = a'_{j+1} = k-1, \qquad k-1 > a'_{j+2} > \cdots > a'_m. \tag{26}$$

Moving the backtrack. From any permutation, $(k-1, k)$ is an edge (Bruhat if the value at $k-1$ is smaller, quantum otherwise), and the pair $(k-1,k)(k-1,k)$ returns to the starting permutation with quantum degree q_{k-1}. Hence the pair can be inserted into or removed from a word at any point without affecting the other edges. For a word with block (26) and $m \geq 3$, let p be the larger of the block entries adjacent to the pair (a'_{j-1} or a'_{j+2}, or the only one present), and move the pair to the other side of p. The result is again a directed path to w with a block of the form (26), with the same $|\omega|$, $D(\omega)$ and $c(\omega)$, and its unique cut differs from j by one, so the two weights cancel. This is an involution without fixed points on the words with a single valid cut and $m \geq 3$.

Surviving words. What remains are the words with empty block, $\omega = \gamma_{>k}(k, b_1)\cdots(k, b_r)$, and the words $\omega = \gamma_{>k}(k-1,k)(k-1,k)(k,b_1)\cdots(k,b_r)$. Inserting or removing the pair is a bijection between the two kinds, the second kind having weight $-\beta^2 q_{k-1}$ times the first (one more chain label, one more left edge, degree q_{k-1}). In both kinds position k is untouched before the right edges, so $c(\omega) = u(k)$. The left side of (25) is therefore

$$\frac{1-\beta^2 q_{k-1}}{1+\beta y_{u(k)}} \sum_{\omega} \beta^{|\omega|} q^{D(\omega)},$$

the sum over directed paths $\omega = \gamma'(k, b_1)\cdots(k, b_r)$ from u to w with γ' a Left-free $(k-1)$-chain without upper index k and $b_1 > \cdots > b_r > k$. By Corollary 28 this sum equals $\sum_\gamma \beta^{s(\gamma)} q^{D(\gamma)}$ over the Left-free k-chains from u to w, which by Corollary 9 is $\beta^{s(u,w;k)} q^{D(u,w;k)}$ if $w \in E(u,k)$ is Left-free and 0 otherwise. This proves (25), hence (24) at level k, and Theorem 22. □

Remark 29. *For $k = 1$ the argument has no block and recovers Proposition 15 at $z = 1/\beta$. The proof shows more precisely that in the junction identity at $a = 0$ the only composites that do not cancel are the sorted composite $\gamma_{>k} \,\big|\, (k, b_1)\cdots(k, b_r)$ of each Left-free endpoint and its backtrack $\gamma_{>k}(k-1,k) \,\big|\, (k-1,k)(k,b_1)\cdots(k,b_r)$. This is the pattern observed numerically for the moved case in Section 4.11 below; the general case differs in that cut-sliding leaves residues proportional to a (Lemma 31) and in that chains with* Left $\neq \varnothing$ *contribute.*

Remark 30 (The specialization of Maeno–Naito–Sagaki)**.** *A second one-point specialization of Theorem 2 is in the literature. Let*

$$u = (n{-}k{+}1,\ n{-}k{+}2,\ \ldots,\ n{+}1,\ \ n{-}k,\ \ldots,\ 2,\ 1) \in S_{n+1},$$

so that positions $1, \ldots, k+1$ carry the $k+1$ largest values in increasing order and the remaining values follow in decreasing order. The admissible k-chains from u are exactly

$$\gamma_j = (k, k{+}1),\ (k{-}1, k{+}1),\ \ldots,\ (j, k{+}1), \qquad 1 \leq j \leq k+1,$$

with γ_{k+1} the empty chain. Indeed, a first label (a,b) with $b \geq k+2$ is neither Bruhat, as $u(a) > u(b)$, nor quantum, as the value $n+1$ at position $k+1$ is not in $(u(b), u(a))$; so all labels have upper index $k+1$, and after γ_j the only label $(a, k+1)$ with $a < j$ that is an edge is $(j-1, k+1)$, because the value at a is smaller than the value $n-k+j$ at $k+1$, excluding a quantum edge, while the values at positions $a+1, \ldots, j-1$ lie between them unless $a = j-1$. For $j \geq 2$ position 1 is untouched, so $u(1) \in$ Fix and the coefficient carries the factor $z + \overline{y}_{u(1)} = (1 - a\,t_{u(1)})/(\beta t_{u(1)})$, $t_c = 1 + \beta y_c$, which vanishes at $a = t_{u(1)}^{-1}$. For $j = 1$ the endpoint is

$$w = (n{-}k{+}2,\ \ldots,\ n{+}1,\ \ n{-}k{+}1,\ \ n{-}k,\ \ldots,\ 1),$$

with Left $= \{u(2), \ldots, u(k)\}$, Out $= \{u(1)\}$, $s = k$, $D = 0$, *and coefficient $a^{k-1} t_{u(1)}^{-1} = t_{u(1)}^{-k}$ at $a = t_{u(1)}^{-1}$. Since $\beta^k B_k^q = \sum_p (-a)^{k-p} F_p^k$, Theorem 2 predicts*

$$\sum_{p=0}^{k} (-1)^p\, t_{u(1)}^{p}\, F_p^k\, \mathfrak{G}_u^q = \mathfrak{G}_w^q \qquad (\beta = -1),$$

which is [4, Proposition 2.3] transported to $QK_T(Fl_{n+1})$ as in their Section 3.1 (with $1 - y_c = \mathbf{e}^{-\epsilon_c}$, $u = w_{k+1}$ and $w = w_k$ in their notation), the key step in their proof that quantum double Grothendieck polynomials represent Schubert classes. Their proof of this special case expands each term of F_p^k by the general Chevalley formula for the weight $w_0 \epsilon_J$, $J \subseteq \{1, \ldots, k\}$.

4.10 The cancellation mechanism: cut composites and local moves

Every term of (21) is a *cut composite*: a directed path in the quantum Bruhat graph

$$u = v_0 \to v_1 \to \cdots \to v_r = v \;\big|\; v \to \cdots \to w,$$

whose first part is an admissible chain at level $k-1$ or $k-2$ and whose second part is a line path at k; the bar marks the cut. The term attached to it is the product of the chain coefficient $C_{u,v}^{(j)}$ (times $a\beta q_{k-1}$ when $j = k-2$) and the line weight $(-\beta)^s \beta^t q^D (1 + \beta y_{v(\pi)(k)})^{-1}$. The whole edge sequence from u to w determines q^D for the chain and the path together, and it determines $v(\pi)$; only the position of the cut and the level distinguish the bookkeeping. The cancellation in (21) is realized by two moves on cut composites, which we now make precise, together with the identity $\beta(z + \overline{y}_c) + a = (1 + \beta y_c)^{-1}$ used in (22).

Lemma 31 (Cut sliding). *Let γ be an admissible $(k-1)$-chain from u to v, let $\pi = (k-1, k) \cdot \pi'$ be a line path at k from v whose first edge is $(k-1, k)$, and put $v' = v t_{k-1,k}$ and $\kappa \in \{1, q_{k-1}\}$ for the weight of the edge $v \to v'$. Then:*

1. *$\gamma' = \gamma \cdot (k-1, k)$ is an admissible $(k-1)$-chain from u to v' if and only if $(k-1, k) \notin \gamma$, and π' is a line path at k from v' with $v(\pi') = v(\pi)$, $s(\pi') = s(\pi) - 1$, $t(\pi') = t(\pi)$ and $q^{D(\pi)} = \kappa\, q^{D(\pi')}$.*

2. *If $v(k-1) \neq u(k-1)$ and $v(k) \neq u(k-1)$, then* Fix, Left, Out *computed for v and for v' agree, so that $C_{u,v'}^{(k-1)} = \beta\kappa\, C_{u,v}^{(k-1)}$, and the two terms of (21) indexed by (v, π) and (v', π') cancel exactly.*

3. *If $v(k-1) = u(k-1)$, then γ is also an admissible $(k-2)$-chain to v, $C_{u,v}^{(k-1)} = (z + \overline{y}_{u(k-1)}) C_{u,v}^{(k-2)}$, and the two terms indexed by (v, π) and (v', π') sum to*

$$a\kappa\, (-\beta)^{s(\pi')} \beta^{t(\pi')} q^{D(\pi')}\, \frac{C_{u,v}^{(k-2)}}{1 + \beta y_{v(\pi)(k)}}. \tag{27}$$

Proof. (1) The edge $(k-1,k)$ joins adjacent positions, so it is always an edge of the quantum Bruhat graph, Bruhat if $v(k-1) < v(k)$ and quantum with weight q_{k-1} otherwise; it is the same edge in γ' and in π, whence the statement about weights. In γ' the second entries remain weakly decreasing since k is the least possible value, and the only condition of Definition 1 that involves the new last label is distinctness; condition 4 for the previous label (a_r, b_r) is automatic, because $b_r > k$, or $b_r = k$ and $a_r \neq k-1$. Removing the first edge of a line path leaves a line path with the same left-part endpoint $v(\pi)$.

(2) Passing from v to v' exchanges the values $c = v(k-1)$ and $d = v(k)$. The value c leaves the window $\{1, \ldots, k-1\}$: it cannot lie in Left (a value in position $k-1$ cannot have moved left), and if it lies in Out or is not a window value of u at all, its status is unchanged. The value d enters position $k-1$: if $d = u(i)$ with $i \leq k-2$ it was in Out and remains there, and otherwise it is not a window value. Hence m is unchanged while the chain length grows by one and the quantum weight is multiplied by κ, giving $C^{(k-1)}_{u,v'} = \beta\kappa C^{(k-1)}_{u,v}$. The line weights of π and π' differ by the factor $(-\beta)\kappa$, so the two terms are negatives of each other.

(3) Here $v(k-1) = u(k-1)$, so by Corollary 9(2) position $k-1$ is untouched by γ: no label of γ has first entry $k-1$, and γ is admissible at level $k-2$ with the same data; this gives the factorization of $C^{(k-1)}_{u,v}$ as in (22). In v' the value $c = u(k-1)$ has moved to position k, so it passes from Fix to Out while d behaves as in (2); thus $C^{(k-1)}_{u,v'} = \kappa\,(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1} C^{(k-1)}_{u,v}$. Writing $X = (-\beta)^{s(\pi')}\beta^{t(\pi')} q^{D(\pi')}(1+\beta y_{v(\pi)(k)})^{-1}$, the term of (v,π) is $X(-\beta)\kappa C^{(k-1)}_{u,v}$ and the term of (v',π') is $X\kappa(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1} C^{(k-1)}_{u,v}$. Their sum is

$$X\kappa\,\frac{(1+\beta y_c)^{-1} - \beta(z+\overline{y}_c)}{z+\overline{y}_c}\, C^{(k-1)}_{u,v} = X\kappa a\, C^{(k-2)}_{u,v},$$

which is (27). □

The case $v(k) = u(k-1)$, in which a fixed window value would enter position $k-1$, does not occur in a cut composite whose endpoint satisfies $w(k) = u(k)$: it did not appear in any of the verified ranges below, and we do not use it. The residue (27) has exactly the shape of a level-$(k-2)$ term of (21), up to the factor βq_{k-1}: the missing q_{k-1} appears either through $\kappa = q_{k-1}$ or through a factor q_{k-1} inside $q^{D(\pi')}$, and the missing β through the line path π' being one edge shorter. This is why the junction input is $P_{k-1}(u) + a\beta q_{k-1} P_{k-2}(u)$ and not $P_{k-1}(u)$ alone.

Lemma 32 (Braid move)**.** *Let $a < k-1$. For any permutation v, the two-edge sequences $(k-1,k),(a,k)$ and $(a,k-1),(k-1,k)$ from v have the same endpoint $vt_{k-1,k}t_{ak} = vt_{a,k-1}t_{k-1,k}$, and if both are directed paths in the quantum Bruhat graph they have the same quantum weight. Consequently, if γ is an admissible $(k-2)$-chain to v with $\gamma\cdot(a,k-1)$ admissible, and $\pi = (k-1,k)\cdot(a,k)\cdot\pi''$ is a line path from v, then $(k-1,k)\cdot\pi''$ is a line path from $vt_{a,k-1}$ with the same $v(\pi)$, and the terms indexed by (v,π) and $(vt_{a,k-1},(k-1,k)\cdot\pi'')$ at level $k-2$ cancel exactly whenever the window statistics of v and $vt_{a,k-1}$ agree.*

Proof. The identity $t_{k-1,k}t_{ak} = t_{a,k-1}t_{k-1,k}$ holds in S_∞. For the weights write $x = v(a)$, $y = v(k-1)$, $z = v(k)$. The adjacent transposition $(k-1,k)$ is always an edge, quantum exactly when the larger value is on the left. If $x < y$, the edge (a,k) from $vt_{k-1,k}$ is Bruhat, and its cover condition applied to the intermediate position $k-1$, which holds z, gives $z \notin (x,y)$; hence $[y > z] = [x > z]$, and both paths have weight $q_{k-1}^{[y>z]}$. If $x > y$, the edge (a,k) from $vt_{k-1,k}$ is quantum, so $y < z < x$, and both paths have weight $q_a q_{a+1}\cdots q_{k-1}$. The label $(a,k-1)$ is admissible at level $k-2$ since $a \leq k-2 < k-1$, and second entries remain weakly decreasing because $k-1$ is the least possible

value. The chain gains one edge (a factor β when m is unchanged) while the line path loses one left edge (a factor $-\beta$), and the quantum weights agree, so the two terms cancel. $\square$

When the window statistics change in Lemma 32 (the value $v(a)$ or $v(k-1)$ is a fixed window value), a residue of the same kind as (27) is produced instead.

The drop term. The diagonal term of (21), $\pi = \varnothing$ and $v = w$, contributes $(1+\beta y_{w(k)})^{-1}\big(C^{(k-1)}_{u,w} + a\beta q_{k-1} C^{(k-2)}_{u,w}\big)$, and after (22) what is left of it is the drop term in (23). In the language of cut composites it is the composite with an empty line path, and Lemma 31 pairs it with the composite $u \xrightarrow{\gamma_w} w \xrightarrow{(k-1,k)} wt_{k-1,k} \;\big|\; (k-1,k)$, for which the move of (1) is blocked because the label $(k-1,k)$ would be repeated. Its term is $-\beta\kappa'\,(1+\beta y_{u(k)})^{-1} C^{(k-1)}_{u,wt_{k-1,k}}$ with $\kappa\kappa' = q_{k-1}$, and by the computation in the proof of Lemma 31 it equals $-\beta^2 q_{k-1}(1+\beta y_{u(k)})^{-1} C^{(k-1)}_{u,w}$ when $w(k-1) \neq u(k-1)$, and $-\beta q_{k-1}(1+\beta y_{u(k)})^{-1}(1+\beta y_{u(k-1)})^{-1} C^{(k-2)}_{u,w} = -\beta q_{k-1}(1+\beta y_{u(k)})^{-1}\big(\beta C^{(k-1)}_{u,w} + a C^{(k-2)}_{u,w}\big)$ when $w(k-1) = u(k-1)$, using (22) at level $k-1$. In both cases it is the negative of the drop term, provided $C^{(k-2)}_{u,w} = 0$ in the first case.

A worked class. Take $u = 21345$, $k = 3$ and the endpoint $w = 12345$, which is not in $E(u,3)$, so the right side of (21) must vanish. The nonzero terms are indexed by three cut composites of the single edge sequence $(1,3),(2,3),(1,3)$ from u to w at level 2, its braid image at level 1, and the drop:

$$\begin{array}{lll}
\text{level } 2: & (1,3)\,\big|\,(2,3),(1,3) & \beta^2 q_1 q_2\,(z+\overline{y}_1)\,t_2^{-1}t_3^{-1},\\
\text{level } 2: & (1,3),(2,3)\,\big|\,(1,3) & -\beta q_1 q_2\,t_1^{-1}t_2^{-1}t_3^{-1},\\
\text{level } 1: & (1,3)\,\big|\,(2,3),(1,3) & a\beta^3 q_1 q_2^2\,t_2^{-1}t_3^{-1},\\
\text{level } 1: & (1,3),(1,2)\,\big|\,(2,3) & -a\beta^3 q_1 q_2^2\,t_2^{-1}t_3^{-1},\\
\text{drop}: & & a\beta q_1 q_2\,t_2^{-1}t_3^{-1},
\end{array}$$

with $t_c = 1+\beta y_c$. The two level-1 terms are a braid pair (Lemma 32) and cancel. The two level-2 terms are a cut sliding pair of type (3) in Lemma 31, since the slid value $v(2) = 1 = u(2)$ is fixed; their sum is the residue $-a\beta q_1 q_2 t_2^{-1}t_3^{-1}$, which cancels the drop term. The residue is absorbed by the drop rather than by a level-1 path term because the relevant level-1 endpoint is w itself.

Status. Lemmas 31 and 32 are proved. What remains for the diagonal case is the global statement: the cut composites with a given endpoint w, $w(k) = u(k)$, decompose into classes generated by the two moves, and within each class the exact cancellations and the residues of types (27) sum to zero against the level-$(k-2)$ terms and the drop. We have verified this termwise, with the symbolic coefficients, for all $u \in S_5$ and all k, and for a sample of $u \in S_6$ with $k \leq 5$; the identity (23) holds for every endpoint w in these ranges, including endpoints outside $E(u,k)$, where it asserts the vanishing of the right side. A proof of the global statement, and of its analogue for the moved cases below, would complete the induction.

The moved cases. Suppose $u(k) \neq w(k)$. Then $C^{(k-1)}_{u,w}$ and $C^{(k-2)}_{u,w}$ need not vanish, in contrast to the quantum Schubert case: a $(k-1)$-chain may move position k through a label (a,k). The identity (21) must then be established by matching each k-chain to w with a $(k-1)$- or $(k-2)$-chain to some v followed by a line path $v \to w$, and by cancelling all remaining terms in pairs. The $k = 1$ case shows the pattern: there $P_{k-2} = 0$, every nonempty line path is a single up-path, and the matching is the identity. The complete case analysis for $k \geq 2$ is not carried out in this version.

4.11 Special cases

Two specializations isolate the two difficulties of the junction identity from each other.

The classical case $q = 0$. Set all $q_j = 0$. Then all edges are Bruhat covers, the level-$(k-2)$ part of the junction input and the drop term vanish (both carry q_{k-1}), and (20) reduces to

$$\beta\, P_k(u) = \mathcal{T}_k P_{k-1}(u) - a\, P_{k-1}(u), \tag{28}$$

the expansion of $\beta B_k + aB_{k-1} = (1+\beta x_k)B_{k-1}$ by the classical equivariant K-Chevalley formula. The diagonal case is then vacuous:

Lemma 33. *Let $q = 0$ and $w(k) = u(k)$. There is no admissible $(k-1)$-chain γ from u to some v together with a nonempty line path at k from v to w. Consequently* (28) *holds at every such w, by* (22).

Proof. Follow the value in position k along the composite path. Every label (a,k), whether in γ or in the left part of the line path, is a Bruhat cover, so it replaces the value in position k by a strictly smaller one; every right edge (k,b) replaces it by a strictly larger value coming from a position $b > k$. Suppose first that some label (a,k) occurs, and let (α,k) be the first. It moves $u(k)$ to position α, which is touched afterwards only by a left edge (α,k) of the line path; that edge would have to be a Bruhat cover with $u(k) = v(\alpha) < v(k)$, whereas every value occupying position k after (α,k) is smaller than $u(k)$. Hence $u(k)$ never returns to position k, and the right edges, which only import values from positions beyond k, cannot return it either. If no label (a,k) occurs, the line path consists of right edges only, and the first of them moves $u(k)$ to a position $b > k$ that is never touched again. In all cases $w(k) \neq u(k)$. □

For the moved case at $q = 0$ the computations of Section 4.10 show that only cut sliding is needed, together with the end braid $t_{k-1,k}t_{kb} = t_{k-1,b}t_{k-1,k}$, which converts a right edge following a junction edge $(k-1,k)$ into two $(k-1)$-chain labels; the residues (27) cancel the term $-aC^{(k-1)}_{u,w}$ of (21) exactly as in the example $u = 21435$, $k = 3$, $w = 24315$, where the pair $u \,\big|\, (2,3),(3,4)$ and $ut_{23} \,\big|\, (3,4)$ leaves the residue $a\beta(z+\overline{y}_2)t_1^{-1} = aC^{(2)}_{u,w}$.

The word-and-cut bookkeeping of Section 4.9 organizes the classical case completely at the level of individual words. We normalize the coefficients by

$$\widehat{C}^{(j)}_{u,v} := \beta^j C^{(j)}_{u,v} = \beta^s \prod_{i=1}^{j} \phi_i(u,v;j), \qquad \phi_i = \begin{cases} t^{-1}_{u(i)} - a & u(i) \in \mathrm{Fix}, \\ a & u(i) \in \mathrm{Left}, \\ t^{-1}_{u(i)} & u(i) \in \mathrm{Out}, \end{cases}$$

with $t_c = 1 + \beta y_c$; this is (6) at $q = 0$ written with $z + \overline{y}_c = (t_c^{-1} - a)/\beta$ and $m + |\,\mathrm{Fix}\,| = j$. In this notation the classical junction identity (28) at the endpoint w reads

$$\widehat{C}^{(k)}_{u,w} + a\,\widehat{C}^{(k-1)}_{u,w} = \sum_{(\gamma,\pi)\colon u\to w} \beta^{s(\gamma)}(-\beta)^{s(\pi)}\beta^{t(\pi)}\, t^{-1}_{v(\pi)(k)} \prod_{i<k} \phi_i(u, \mathrm{end}\,\gamma; k-1), \tag{29}$$

the sum over composites of an admissible $(k-1)$-chain γ and a line path π of Bruhat edges. As in Section 4.9, a composite has a word $\omega = \gamma_{>k}\,(a'_1,k)\cdots(a'_m,k)\,(k,b_1)\cdots(k,b_r)$ and a cut j; now the chain part may contain descents, so the valid cuts are no longer determined by monotonicity alone. Call the block entry a'_r *fresh* if position a'_r is touched neither by $\gamma_{>k}$ nor by $a'_1,\dots,a'_{r-1}$, and write $\Sigma(\omega)$ for the sum of the weights of all composites with word ω.

Proposition 34 (Classical word lemma). *Let $q = 0$ and let ω be the word of a composite, with block $(a'_1, \dots, a'_m)$, and let S be the length of the longest strictly decreasing suffix $a'_{m-S+1} > \cdots > a'_m$ of the block ($S = 0$ if $m = 0$). Then:*

1. *The block entries are pairwise distinct.*
2. *The valid cuts form the interval $m - S \le j \le J$, where J is the least index $r < m$ such that a'_r is not fresh and $a'_r > a'_{r+1}$, or $J = m$ if there is no such r.*
3. *$\Sigma(\omega) \neq 0$ if and only if $J = m$ and all entries $a'_{m-S+1}, \dots, a'_m$ of the decreasing suffix are fresh. In that case*
$$\Sigma(\omega) = \beta^{|\omega|}\, t_{c(\omega)}^{-1}\, a^S \prod_{i \notin \{a'_1, \dots, a'_m\}} \phi_i^0 \prod_{r \le m-S} \psi_r,$$
where $c(\omega)$ is the value at position k after the block, $\phi_i^0 = \phi_i(u, u\gamma_{>k}; k-1)$, and for $r \le m-S$: $\psi_r = \phi_{a'_r}^0$ if a'_r is not fresh, $\psi_r = a$ if a'_r is fresh and $a'_r > a'_{r+1}$, and $\psi_r = t_{u(a'_r)}^{-1}$ if a'_r is fresh and either $a'_r < a'_{r+1}$ or $r = m - S$.

Proof. (1) A Bruhat edge (a, k) requires the value at a to be smaller than the value at k, and it moves the smaller value to k; so along the block the value at position k strictly decreases. If (a, k) occurred at two block indices $r < r'$, the value placed at a by the first occurrence would still be there at the second, and it exceeds every later value at k; so the second occurrence is not a Bruhat edge.

(2) By (1), an entry a'_r is fresh exactly when position a'_r is not touched by $\gamma_{>k}$. The line part $(a'_{j+1}, k) \cdots (a'_m, k)(k, b_1) \cdots (k, b_r)$ is a line path exactly when $a'_{j+1} > \cdots > a'_m$, that is $j \ge m - S$. The chain part $\gamma_{>k}(a'_1, k) \cdots (a'_j, k)$ has distinct labels by (1) and weakly decreasing upper indices, and condition 4 of Definition 1 is automatic inside $\gamma_{>k}$ and at its last label. At a block index $r < j$ the lower index a'_r is repeated exactly when a'_r is not fresh, since at level $k - 1$ a window position is touched only by labels with that lower index, and then condition 4 requires $(a'_r, k) \prec (a'_{r+1}, k)$, that is $a'_r < a'_{r+1}$. Hence the chain part is admissible exactly when $j \le J$.

(3) Fix a valid cut j and let v_j be the endpoint of the chain part. We determine the status of each window position $i < k$ in v_j, starting from its status in $v_0 = u\gamma_{>k}$. Let x_r be the value at a'_r just before (a'_r, k); after this label x_r is at k, and after (a'_{r+1}, k) (if $r < j$) it is at a'_{r+1}, where it stays, since by (1) position a'_{r+1} is not touched again by the block. If a'_r is fresh then $x_r = u(a'_r)$, so position a'_r has status Out if $r = j$, status Left if $r < j$ and $a'_{r+1} < a'_r$, and status Out if $r < j$ and $a'_{r+1} > a'_r$; before the label, for $r > j$, its status is Fix. If a'_r is not fresh, then $x_r \neq u(a'_r)$, and we claim that the status of every value is unchanged by the block labels of the chain part other than the fresh ones just described. Indeed, a value $c = u(i)$ with status Left in v_0 sits at a position $p < i$ reached by a descent $(i, b), (p, b)$ of $\gamma_{>k}$, and a chain part containing (p, k) would contain the subsequence $(i, b), (p, b), (p, k)$ with $p < i \le k - 1 < k < b$, which the prohibition lemma [2, Lemma 2.9] excludes for an admissible chain. A value with status Out at a window position p is moved by (p, k) to position k, and by the next label, if any, to a'_{r+1} with $a'_{r+1} > p$ by condition 4, since p is a repeated lower index; so it keeps status Out. Values at positions greater than k are not moved by the block. This proves that the weight of the composite with cut j is
$$\beta^{|\omega|}(-1)^{m-j}\, t_{c(\omega)}^{-1} \prod_{i \notin \{a'_r\}} \phi_i^0 \prod_{a'_r \text{ not fresh}} \phi_{a'_r}^0 \prod_{a'_r \text{ fresh}} \phi_r^{(j)},$$

where

$$\phi_r^{(j)} = \begin{cases} a & r < j,\ a'_{r+1} < a'_r, \\ t^{-1}_{u(a'_r)} & r < j,\ a'_{r+1} > a'_r, \text{ or } r = j, \\ t^{-1}_{u(a'_r)} - a & r > j, \end{cases}$$

the sign coming from the $m - j$ left edges of the line part.

Now sum over $j \in [m - S, J]$. If $J < m - S$ there is no valid cut and $\Sigma(\omega) = 0$; note that $J \neq m - S$ when $m - S \geq 1$, since a'_{m-S} is an ascent by the maximality of S. For $r < m - S$ we have $r < j$ throughout, so $\phi_r^{(j)} = \psi_r$ is independent of j; for $r = m - S \geq 1$, the entry a'_{m-S} is an ascent, so $\phi^{(j)}_{m-S} = t^{-1}_{u(a'_{m-S})}$ both for $j = m - S$ and for $j > m - S$. If $m - S < J < m$, the entries a'_r with $m - S < r < J$ are fresh descents, a'_J is not fresh, and the factors for $r > J$ do not depend on j. Writing $\tau_i = t^{-1}_{u(a'_{m-S+i})}$ and $T_l = a^l \prod_{l<i<L+1}(\tau_i - a)$ for $0 \leq l \leq L := J - (m - S) - 1$, the remaining sum is, up to a common factor,

$$\sum_{j=m-S}^{J} (-1)^{m-j} \prod_{m-S<r<J} \phi_r^{(j)} \\ = \pm\Big(T_0 - (T_0 + T_1) + (T_1 + T_2) - \cdots + (-1)^L (T_{L-1} + T_L) + (-1)^{L+1} T_L\Big) = 0,$$

because the cut $j = m - S + l$ with $1 \leq l \leq L$ contributes $a^{l-1}\tau_l \prod_{i>l}(\tau_i - a) = T_{l-1} + T_l$, the cut $j = m - S$ contributes T_0, and the cut $j = J$ contributes $a^L = T_L$. If $J = m$ and some suffix entry is not fresh, it must be a'_m (the others are descents), its factor $\phi^0_{a'_m}$ is constant, and the same telescoping with $L = S - 1$ gives 0. If $J = m$ and all suffix entries are fresh, the same computation with $L = S$, where now the cut $j = m$ contributes $a^{S-1}\tau_S = T_{S-1} + T_S$, gives $T_S = a^S$. Collecting the constant factors yields the stated formula. □

Thus at $q = 0$ a word contributes either nothing or a single product. In the surviving words every entry of the decreasing suffix of the block is a fresh position whose value, after the block, has moved to the left (status Left at level k), and the suffix contributes exactly the factor a^S of (6) for these values. What remains for the classical case is the matching: for every w, the words with $\Sigma(\omega) \neq 0$ and endpoint w must have total weight $\widehat{C}^{(k)}_{u,w} + a\widehat{C}^{(k-1)}_{u,w}$. We have verified for all $u \in S_5$ and all $k \leq 5$ that there is in fact *at most one* such word for each w, and that its $\Sigma(\omega)$ equals the right side; in the examples the surviving word is obtained from the k-chain, or from the $(k-1)$-chain when $w \in E(u, k-1)$, by commutations of disjoint transpositions as in Lemma 25 and braid relations such as $t_{kb}t_{ab} = t_{ak}t_{kb}$ and $t_{ab}t_{ak} = t_{ak}t_{kb}$, which convert labels into junction labels (a, k) and right edges (k, b). We have not proved this matching.

There is also a second route in the classical case, which has no quantum analogue. Since $\prod_{j\leq k}(x_j + z)$ is symmetric in x_i, x_{i+1} for $i \neq k$, the isobaric divided difference π_i^x commutes with multiplication by it, and $\pi_i^x \mathfrak{G}_u = \mathfrak{G}_{us_i}$ when $us_i < u$, $\pi_i^x \mathfrak{G}_w = -\beta \mathfrak{G}_w$ when $ws_i > w$. Hence the classical coefficients must satisfy

$$C_{us_i,w'} = \begin{cases} C_{u,w's_i} - \beta\, C_{u,w'} & w's_i > w', \\ 0 & w's_i < w', \end{cases} \qquad (us_i < u,\ i \neq k,\ q = 0). \tag{30}$$

We have verified that the closed form (6) at $q = 0$ satisfies (30) for all $u \in S_4$ and all k (1200 instances). The identity is suggestive: when a window value c moves from position $i+1$ to position i, the two terms on the right combine as $(1 + \beta y_c)^{-1} - \beta(z + \overline{y}_c) = 1 - \beta z$, which is the origin of the

factor $(1-\beta z)^{|\,\mathrm{Left}\,|}$. Granting (30) for the chain combinatorics, the classical theorem for all u follows by descending induction in the right weak order from the case of k-Grassmannian u (no descents outside position k). For such u the product is symmetric in $x_1,\dots,x_k$ and in the remaining variables, all endpoints are again k-Grassmannian (checked for all $u \in S_6$), and the statement is an equivariant K-theoretic Pieri rule for the Grassmannian, multiplying by $\prod_{j\le k}(x_j+z) = (1-\beta z)^k \prod_j (x_j \oplus z')$ with $z' = z/(1-\beta z)$; the expansion of this product into the classes $\mathfrak{G}_{c[k,p]}$ is the case $u = e$ of Theorem 2.

The case $k = 2$. For $k = 2$ a line path has at most one left edge, namely $(1,2)$, and the level-0 part of the junction input is $a\beta q_1 \mathfrak{G}_u$. The cut composites are the sequences

$$(1,b_1),\dots,(1,b_r) \;\big|\; [(1,2)]\; (2,c_1),\dots,(2,c_t),$$

with the junction edge $(1,2)$ possibly absent, present on one side of the cut, or present on both sides. The last possibility is a *backtrack*: a chain ending with a quantum edge $(1,2)$ followed by the same edge reversed as a Bruhat line edge, or vice versa. Its contribution relative to the composite without the two edges is $\beta\cdot(-\beta)\cdot q_1 = -\beta^2 q_1$, and this is exactly how the factor $1-\beta^2 q_1$ on the left of (20) is produced. For instance, for $u = 21345$ and $w = 32145 \in E(u,2)$, the only terms are $ut_{13} \,\big|\, (2,3)$, with value $\beta t_1^{-1} t_2^{-1}$, and the backtrack $ut_{13}t_{12} \,\big|\, (1,2),(2,3)$, with value $-\beta^3 q_1 t_1^{-1} t_2^{-1}$, whose sum is $\beta(1-\beta^2 q_1) C^{(2)}_{u,w}$. For $w = 23145$ the backtrack $ut_{12} \,\big|\, (1,2),(2,3)$ has value $-\beta^2 q_1 t_1^{-1} t_2^{-1}$ and the level-0 composite $u \,\big|\, (2,3)$ has value $a\beta^2 q_1 t_1^{-1}$; their sum is $-\beta^3 q_1 (z+\overline{y}_2) t_1^{-1} = -\beta^2 q_1\, \beta C^{(2)}_{u,w}$, again by $(1+\beta y_2)^{-1} - a = \beta(z+\overline{y}_2)$. Thus for $k = 2$ all three mechanisms—cut sliding, backtracks, and the absorption of residues by the level-0 terms—are visible, while the braid move of Lemma 32 cannot occur. We have verified (21) for $k = 2$ termwise for all $u \in S_5$. The diagonal case can be settled completely.

Proposition 35 (Diagonal case for $k = 2$)**.** *Let $k = 2$ and $w(2) = u(2)$. Then (23) holds; hence (21) holds at w.*

Proof. We classify the nonempty cut composites ending at w. Position 2 is touched by every edge of a line path at 2. A right edge $(2,c)$ sends the value in position 2 to the position $c > 2$, which is never touched again, so if the line path contains a right edge, the final value in position 2 is not $u(2)$ unless $u(2)$ was brought to position c earlier—impossible, as c is touched only once. Hence the line path is $\pi = (1,2)$, and the value $u(2)$ sits in position 1 of v, so the chain γ to v contains the label $(1,2)$. For a level-1 chain this label is the last one, as 2 is the least admissible second entry, and $(1,2)$ moves $u(2)$ to position 1 only if it is the first label whose second entry is 2; since labels are distinct, $\gamma = \gamma_0 \cdot (1,2)$ with $(1,2) \notin \gamma_0$, and $w = vt_{12} = u\gamma_0$. Thus γ_0 is the unique 1-chain from u to w (Proposition 15), and there is exactly one level-1 composite, the backtrack $\gamma_w \cdot (1,2) \,\big|\, (1,2)$. There is no level-0 composite, because a line path from u cannot restore $u(2)$ either. Write κ, κ' for the weights of the two traversals of $(1,2)$, so $\kappa\kappa' = q_1$.

If $w \neq u$, the window value $u(1)$ is in Out for both w and wt_{12} (it has left position 1 and $w(1) \neq u(1)$), so by Lemma 31(2) $C^{(1)}_{u,wt_{12}} = \beta\kappa C^{(1)}_{u,w}$, and the backtrack contributes $(-\beta)\kappa'(1+\beta y_{u(2)})^{-1}\beta\kappa C^{(1)}_{u,w} = -\beta^2 q_1 (1+\beta y_{u(2)})^{-1} C^{(1)}_{u,w}$, while the drop term is $\beta q_1 (1+\beta y_{u(2)})^{-1} \beta C^{(1)}_{u,w}$ because $C^{(0)}_{u,w} = 0$. They cancel.

If $w = u$, then γ_0 is empty, $C^{(1)}_{u,u} = z + \overline{y}_{u(1)}$, and $u(1)$ passes from Fix to Out in ut_{12}, so $C^{(1)}_{u,ut_{12}} = \kappa(1+\beta y_{u(1)})^{-1}$ and the backtrack contributes $-\beta q_1 (1+\beta y_{u(1)})^{-1}(1+\beta y_{u(2)})^{-1}$. The drop term is $\beta q_1 (1+\beta y_{u(2)})^{-1}\big(\beta(z+\overline{y}_{u(1)}) + a\big) = \beta q_1 (1+\beta y_{u(1)})^{-1}(1+\beta y_{u(2)})^{-1}$. They cancel. □

The moved case for $k = 2$ requires in addition the classification of 2-chains: their labels form segments $(1, b)$, $(2, b)$, $(1, b)(2, b)$ or $(2, b)(1, b)$ for strictly decreasing b, the last form being allowed only at the first occurrence of the lower index 2. The computations through S_5 exhibit the following pattern, which is what a proof must establish. If $w \in E(u, 2)$ and its 2-chain has no segment of the form $(2, b)(1, b)$, then sorting the chain, all labels $(1, b)$ first and all labels $(2, c)$ afterwards, gives a cut composite $\gamma^* \,\big|\, \pi^*$ whose term is $\beta C^{(2)}_{u,w}$, and its backtrack $\gamma^*(1, 2) \,\big|\, (1, 2)\pi^*$ contributes $-\beta^3 q_1 C^{(2)}_{u,w}$; for instance $u = 24135$, $w = 31524$, with chain $(2, 5), (1, 4), (2, 3)$ and composite $(1, 4) \,\big|\, (2, 5), (2, 3)$. A segment $(2, b)(1, b)$ is where a window value moves left, and it is resolved by $t_{2b}t_{1b} = t_{1b}t_{12}$: for $u = e$ and $w = 231$, with chain $(2, 3), (1, 3)$, the composites are $(1, 2) \,\big|\, (2, 3)$ with term βt_1^{-2} and $\varnothing \,\big|\, (1, 2), (2, 3)$ with term $-\beta^2(z+\overline{y}_1)t_1^{-1}$, a cut sliding pair of type (3) whose residue $\beta a t_1^{-1}$ is exactly $\beta C^{(2)}_{e,231}$: the factor $1-\beta z$ attached to a value in Left is produced by $(1+\beta y_c)^{-1} - \beta(z+\overline{y}_c) = 1-\beta z$. The level-0 composite $e \,\big|\, (1, 2), (2, 3)$, with term $-a\beta^3 q_1 t_1^{-1}$, then cancels the part $\beta^3 q_1 C^{(2)}_{e,231}$ of the left side. When the 1-chain to w ends with $(1, 2)$, the term $-a(1 - \beta^2 q_1)C^{(1)}_{u,w}$ is cancelled by the residue of the pair $u \,\big|\, (1, 2)\pi'$, $ut_{12} \,\big|\, \pi'$ together with the level-0 composite $u \,\big|\, (1, 2)\pi'$, using $t_{12}t_{2c_1} \cdots t_{2c_t} = t_{1c_1} \cdots t_{1c_t}t_{12}$ and Postnikov's lemma for the equality of weights. The outstanding point is the validity in the quantum Bruhat graph of the sorted and braided sequences; Lemma 26 settles the sorting for chains without descents, and Lemma 25 is the local step for moving a right edge past a label with smaller upper index in general, but the braided segments $(2, b)(1, b)$ are not covered, and we have not completed this case analysis.

The y-recursion. Finally, there is a reduction that is valid for all q and does not use the Chevalley formula at all. The quantization map commutes with the isobaric divided differences π_i^y in the second alphabet, and $\pi_i^y \mathfrak{G}^q_w = \mathfrak{G}^q_{s_i w}$ if $s_i w < w$, $\pi_i^y \mathfrak{G}^q_w = -\beta \mathfrak{G}^q_w$ otherwise. Applying π_i^y to (5) for $s_i u > u$ and using $\pi_i(fg) = (1 + \beta y_{i+1})\partial_i(f)g + s_i(f)\pi_i(g)$ shows that the theorem for $s_i u$ implies the theorem for u provided the coefficients satisfy

$$C_{u,w} = \pi_i^y\big(C_{s_i u,w}\big) + \begin{cases} s_i\big(C_{s_i u,\, s_i w}\big) & s_i w > w, \\ \beta\, s_i\big(C_{s_i u,w}\big) & s_i w < w. \end{cases} \tag{31}$$

We have verified (31) for the closed form, with q arbitrary, for all $u \in S_5$ and $k \le 3$. It reduces the theorem for all $u \in S_n$ to the single permutation $u = w_0^{(n)}$, for which the chains are indexed by subsets of $\{1, \ldots, k\}$ in the classical case; the quantum case of this base is not simpler than the general one, which is why we rely on the Chevalley induction instead.

5 A small example and specializations

Example 36. *Take $u = 21$ and $k = 1$. The endpoints are $21, 312, 132$, and the identity permutation e. Their chain data (s, D) are respectively*

$$(0, 0), \qquad (1, 0), \qquad (2, (1, 0, \ldots)), \qquad (1, (1, 0, \ldots)).$$

For instance, the path to 132 is

$$213 \xrightarrow{(1,3)} 312 \xrightarrow{(1,2)} 132,$$

whose second edge is quantum. Since $B_1^q = x_1 + z - \beta q_1(1+\beta x_1)$*, the formula reads*

$$\begin{aligned}\big(x_1 + z - \beta q_1(1+\beta x_1)\big)\mathfrak{G}_{21}^q = &\left(z - \frac{y_2}{1+\beta y_2}\right)\mathfrak{G}_{21}^q \\ &+ \frac{\mathfrak{G}_{312}^q + \beta q_1 \mathfrak{G}_{132}^q + q_1}{1+\beta y_2}.\end{aligned} \tag{32}$$

Here $\mathfrak{G}_e^q = 1$*, and all polynomials have alphabets* $(x; y)$*. This particular identity can be verified by expanding the four quantum double Grothendieck polynomials and clearing the denominator.*

Classical specialization. When all $q_i = 0$, only paths without quantum edges contribute. The multiplier becomes $\prod_{i=1}^{k}(x_i + z)$, giving the classical double Grothendieck top-block formula as a specialization of Theorem 2. In particular, the equivariant classical specialization is itself new, rather than an appeal to a previously published top-block formula.

Non-equivariant specialization. When all $y_a = 0$, the coefficient becomes

$$q^D \beta^{s-m} z^{|\,\mathrm{Fix}\,|}(1-\beta z)^{|\,\mathrm{Left}\,|}.$$

In addition setting $z = 0$ leaves only endpoints with $\mathrm{Fix} = \varnothing$, and the surviving coefficient is $q^D\beta^{s-k}$. At $\beta = -1$ it is $(-1)^{s-k}q^D$. This specialization is Theorem 18, derived from the quantum Grothendieck Pieri formula of Naito–Sagaki [5] through the uniqueness of chains and Lemma 17.

Cohomological specialization. At $\beta = 0$, terms with $s > m$ vanish. Those with $s = m$ have coefficient

$$q^D \prod_{a \in \mathrm{Fix}} (z - y_a).$$

The corresponding comparison is with the top quantum double Schubert block in coefficient alphabet $-y$.

Usual K-theory variables. At $\beta = -1$ the multiplicative equivariant variables in these conventions may be written

$$Y_a = \frac{1}{1-y_a}.$$

Thus comparisons with formulas in the Y variables use $y_a = 1 - Y_a^{-1}$. This change of variables does not alter the permutation indices or the quantum parameters q_i.

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